# Topology II

Lecture notes, Chapters 9–14 (Algebraic Topology)

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# Chapter 9. The Fundamental Group

## 9.1 Homotopy of paths

**Motivation.** A loop around the hole of the punctured plane cannot be shrunk; homotopy makes this precise and turns spaces into groups.

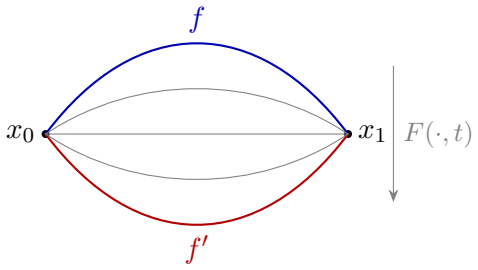



**Figure.** *A path homotopy $F$ from $f$ to $f'$ with end points $x_0$, $x_1$ fixed.*

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ ($I = [0,1]$) with $F(x,0) = f(x)$, $F(x,1) = f'(x)$; $f$ is **nulhomotopic** if it is homotopic to a constant. Paths $f, f' : I \to X$ from $x_0$ to $x_1$ are **path homotopic**, $f \simeq_p f'$, if there is $F : I \times I \to X$ with $F(s,0) = f(s)$, $F(s,1) = f'(s)$, $F(0,t) = x_0$, $F(1,t) = x_1$.

**Lemma 9.1 (51.1)**

$\simeq$ and $\simeq_p$ are equivalence relations.

**Proof.** $F(x,t) = f(x)$ gives $f \simeq f$; if $F : f \simeq f'$, then $F(x, 1-t)$ gives $f' \simeq f$. For $F : f \simeq f'$, $F' : f' \simeq f''$, let $G(x,t) = F(x,2t)$ $(t \le \frac{1}{2})$, $F'(x, 2t-1)$ $(t \ge \frac{1}{2})$, continuous by the pasting lemma. For paths these keep the end points fixed. ■

$[f]$ denotes the path-homotopy class of $f$.

**Example 9.1**

For maps (or paths with the same end points) $f, g$ into a convex subset of $\mathbb{R}^n$, the **straight-line homotopy** $F(x,t) = (1-t)f(x) + tg(x)$ is a homotopy (path homotopy). In $\mathbb{R}^2 - \{0\}$, however, $f(s) = (\cos \pi s, \sin \pi s)$ and $g(s) = (\cos \pi s, -\sin \pi s)$ are not path homotopic (Section 9.4).

For a path $f$ from $x_0$ to $x_1$ and $g$ from $x_1$ to $x_2$, the **product** $f * g$ is $f(2s)$ on $[0, \frac{1}{2}]$ and $g(2s-1)$ on $[\frac{1}{2}, 1]$. If $F : f \simeq_p f'$ and $G : g \simeq_p g'$, then $H(s,t) = F(2s,t)$ $(s \le \frac{1}{2})$, $G(2s-1, t)$ $(s \ge \frac{1}{2})$ is a path homotopy $f * g \simeq_p f' * g'$, so $[f] * [g] = [f * g]$ is defined. For continuous $k : X \to Y$, $k \circ F$ is a path homotopy from $k \circ f$ to $k \circ f'$, and $k \circ (f * g) = (k \circ f) * (k \circ g)$.

**Theorem 9.2 (51.2)**

The operation $*$ on path-homotopy classes satisfies:
(1) (associativity) if $[f] * ([g] * [h])$ is defined, so is $([f] * [g]) * [h]$, and they are equal;
(2) (identities) if $e_x$ is the constant path at $x$ and $f$ runs from $x_0$ to $x_1$, then $[e_{x_0}] * [f] = [f] = [f] * [e_{x_1}]$;
(3) (inverses) with $\bar{f}(s) = f(1-s)$, $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_{x_1}]$.

**Proof.** Let $e_0$ be the constant path in $I$ at 0, $i : I \to I$ the identity path, $\bar{i}$ its reverse.

(2) $e_0 * i \simeq_p i$ in the convex $I$ by some $G$; $f \circ G$ is a path homotopy from $f \circ (e_0 * i) = e_{x_0} * f$ to $f$. The other identity uses $i * e_1$.

(3) $i * \bar{i} \simeq_p e_0$ in $I$ by some $H$; $f \circ H$ gives $f * \bar{f} \simeq_p e_{x_0}$. Similarly $\bar{f} * f \simeq_p e_{x_1}$.

(1) For $0 < a < b < 1$ let $k_{a,b}$ be, on $[0, a]$, $[a, b]$, $[b, 1]$, the positive linear map onto $I$ followed by $f$, $g$, $h$. Then $(f * g) * h = k_{1/2,3/4}$, $f * (g * h) = k_{1/4,1/2}$. For the piecewise linear $\gamma : I \to I$ with $0, a, b, 1 \mapsto 0, c, d, 1$, $k_{c,d} \circ \gamma = k_{a,b}$; as $\gamma \simeq_p i$ in $I$, $k_{a,b} \simeq_p k_{c,d}$. ■

**Theorem 9.3 (51.3)**

Let $f$ be a path in $X$ and $0 = a_0 < a_1 < \cdots < a_n = 1$. Let $f_i$ be the positive linear map of $I$ onto $[a_{i-1}, a_i]$ followed by $f$. Then $[f] = [f_1] * \cdots * [f_n]$.

**Proof.** A bracketing of $f_1 * \cdots * f_n$ is, on the intervals of some subdivision $0 = b_0 < \cdots < b_n = 1$, the linear map onto $I$ followed by $f_i$; so it equals $f \circ \gamma$, $\gamma$ the piecewise linear map of $[b_{i-1}, b_i]$ onto $[a_{i-1}, a_i]$. As $\gamma \simeq_p i$, $f \circ \gamma \simeq_p f$. ■

## 9.2 The fundamental group

**Motivation.** Loop classes at a base point form a group, a homeomorphism invariant.

A **loop** at $x_0$ is a path from $x_0$ to $x_0$. By Theorem 9.2 the classes of loops at $x_0$ form a group under $*$, the **fundamental group** $\pi_1(X, x_0)$; it is trivial for convex $X \subset \mathbb{R}^n$. For a path $\alpha$ from $x_0$ to $x_1$ let $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$, $\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$.

**Theorem 9.4 (52.1)**

$\hat{\alpha}$ is a group isomorphism.

**Proof.** $\hat{\alpha}([f]) * \hat{\alpha}([g]) = [\bar{\alpha}] * [f] * [\alpha] * [\bar{\alpha}] * [g] * [\alpha] = \hat{\alpha}([f] * [g])$. With $\beta = \bar{\alpha}$, $\hat{\beta}(\hat{\alpha}([h])) = [\alpha] * [\bar{\alpha}] * [h] * [\alpha] * [\bar{\alpha}] = [h]$; similarly $\hat{\alpha} \circ \hat{\beta} = \mathrm{id}$. ■

**Corollary 9.5 (52.2)**

If $X$ is path connected and $x_0, x_1 \in X$, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

For the path component $C$ of $x_0$, $\pi_1(C, x_0) = \pi_1(X, x_0)$. A path-connected space with trivial fundamental group is **simply connected**.

**Lemma 9.6 (52.3)**

In a simply connected space, any two paths with the same initial and final points are path homotopic.

**Proof.** For $f, g$ from $x_0$ to $x_1$, $[f] * [\bar{g}] = [e_{x_0}]$; multiply on the right by $[g]$. ■

A continuous $h : (X, x_0) \to (Y, y_0)$ (i.e. $h(x_0) = y_0$) induces the homomorphism $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $h_*([f]) = [h \circ f]$.

**Theorem 9.7 (52.4)**

If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$. If $i$ is the identity of $(X, x_0)$, then $i_*$ is the identity.

**Proof.** $(k \circ h)_*([f]) = [k \circ (h \circ f)] = k_*(h_*([f]))$, and $i_*([f]) = [f]$. ■

**Corollary 9.8 (52.5)**

If $h : (X, x_0) \to (Y, y_0)$ is a homeomorphism, $h_*$ is an isomorphism.

**Proof.** With $k = h^{-1}$, $k_* h_* = (k \circ h)_* = \mathrm{id}$ and $h_* k_* = \mathrm{id}$. ■

## 9.3 Covering spaces

**Motivation.** A covering unwraps loops into paths whose end points detect winding, modelled on $\mathbb{R} \to S^1$.

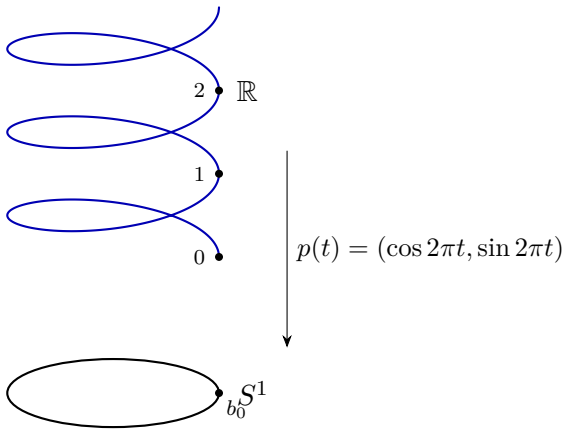



**Figure.** *$p : \mathbb{R} \to S^1$ with fibre $p^{-1}(b_0) = \mathbb{Z}$; a loop at $b_0$ lifts to a path from 0 to its class in $\pi_1(S^1) \cong \mathbb{Z}$.*

Let $p : E \to B$ be continuous and surjective. An open $U \subset B$ is **evenly covered** if $p^{-1}(U)$ is a union of disjoint open $V_\alpha$ (**slices**), each mapped homeomorphically onto $U$. If every point of $B$ has an evenly covered neighbourhood, $p$ is a **covering map** and $E$ a **covering space** of $B$. A covering map is a local homeomorphism and open, and each fibre $p^{-1}(b)$ is discrete.

**Theorem 9.9 (53.1)**

$p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$, is a covering map.

**Proof.** For the open arc $U$ with positive first coordinate, $p^{-1}(U) = \bigcup_n (n - \frac{1}{4}, n + \frac{1}{4})$. $p$ is injective on $[n - \frac{1}{4}, n + \frac{1}{4}]$ (second coordinate increasing), so maps it homeomorphically onto the closed right half circle and $(n - \frac{1}{4}, n + \frac{1}{4})$ onto $U$. Likewise for the arcs with negative first, positive second and negative second coordinate, which with $U$ cover $S^1$. ■

**Example 9.2**

$p|(0, \infty)$ is a surjective local homeomorphism but not a covering map: for any neighbourhood $U$ of $b_0 = (1, 0)$, the component $(0, \epsilon)$ of $p^{-1}(U)$ is not mapped onto $U$.

**Theorem 9.10 (53.2)**

If $p : E \to B$ is a covering map, $B_0 \subset B$ and $E_0 = p^{-1}(B_0)$, then $p_0 = p|E_0 : E_0 \to B_0$ is a covering map.

**Proof.** For $b \in B_0$ and evenly covered $U \ni b$ with slices $V_\alpha$, $p_0^{-1}(U \cap B_0)$ is the disjoint union of the open $V_\alpha \cap E_0$, each mapped homeomorphically onto $U \cap B_0$. ■

**Theorem 9.11 (53.3)**

If $p : E \to B$ and $p' : E' \to B'$ are covering maps, so is $p \times p' : E \times E' \to B \times B'$.

**Proof.** For evenly covered $U \ni b$, $U' \ni b'$ with slices $V_\alpha$, $V'_\beta$, $(p \times p')^{-1}(U \times U')$ is the disjoint union of the open $V_\alpha \times V'_\beta$, each mapped homeomorphically onto $U \times U'$. ■

**Example 9.3**

$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$ covers the torus. Restricting to $b_0 \times S^1 \cup S^1 \times b_0$ (Theorem 9.10) gives a covering of the figure eight by the grid of horizontal and vertical lines through integer points. As $\mathbb{R}_+ \times S^1 \cong \mathbb{R}^2 - 0$ (polar coordinates), $\mathrm{id} \times p : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^2 - 0$ covers the punctured plane.

## 9.4 The fundamental group of the circle

**Motivation.** The covering $\mathbb{R} \to S^1$ gives $\pi_1(S^1) \cong \mathbb{Z}$, on which the rest of the chapter builds.

For $p : E \to B$, a **lifting** of $f : X \to B$ is a map $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

**Lemma 9.12 (54.1)**

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Any path $f : I \to B$ beginning at $b_0$ has a unique lifting $\tilde{f}$ beginning at $e_0$.

**Proof.** By the Lebesgue number lemma choose $0 = s_0 < \cdots < s_n = 1$ with each $f([s_i, s_{i+1}])$ in an evenly covered $U$.

Existence: $\tilde{f}(0) = e_0$; if $\tilde{f}$ is defined on $[0, s_i]$, let $V_0$ be the slice containing $\tilde{f}(s_i)$ and $\tilde{f}(s) = (p|V_0)^{-1}(f(s))$ on $[s_i, s_{i+1}]$ (pasting lemma).

Uniqueness: if another lifting $\tilde{f}'$ equals $\tilde{f}$ on $[0, s_i]$, the connected $\tilde{f}'([s_i, s_{i+1}])$ lies in one slice, $V_0$ as it contains $\tilde{f}(s_i)$; so $\tilde{f}' = (p|V_0)^{-1} \circ f = \tilde{f}$ there. ■

**Lemma 9.13 (54.2)**

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and $F : I \times I \to B$ continuous with $F(0,0) = b_0$. There is a unique lifting $\tilde{F} : I \times I \to E$ with $\tilde{F}(0,0) = e_0$. If $F$ is a path homotopy, so is $\tilde{F}$.

**Proof.** Existence: choose subdivisions $s_0 < \cdots < s_m$, $t_0 < \cdots < t_n$ with each $I_i \times J_j = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ mapped into an evenly covered set (Lebesgue). Lift on $A = 0 \times I \cup I \times 0$ by Lemma 9.12, then over the rectangles in the order $I_1 \times J_1, \ldots, I_m \times J_1, I_1 \times J_2, \ldots$. For the next rectangle $R$, $F(R) \subset U$ evenly covered, and $C = R \cap A'$ ($A' = A$ plus the rectangles done) is its connected left

and bottom edge; $\tilde{F}(C)$ lies in one slice $V_0$, and $\tilde{F} = (p|V_0)^{-1} \circ F$ on $R$ agrees on $C$ and is continuous on $A' \cup R$ (pasting lemma). Uniqueness: a lifting is determined on $A$ (Lemma 9.12), and on each $R$ its connected image meets $V_0$, so lies in $V_0$ and equals $(p|V_0)^{-1} \circ F$. If $F$ is a path homotopy, $\tilde{F}(0 \times I)$ is a connected subset of the discrete fibre $p^{-1}(b_0)$, a point; likewise for $1 \times I$. ■

**Theorem 9.14 (54.3)**

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f, g$ paths in $B$ from $b_0$ to $b_1$ with liftings $\tilde{f}, \tilde{g}$ beginning at $e_0$. If $f \simeq_p g$, then $\tilde{f}$ and $\tilde{g}$ end at the same point and $\tilde{f} \simeq_p \tilde{g}$.

**Proof.** Lift $F : f \simeq_p g$ to $\tilde{F}$ from $e_0$, a path homotopy (Lemma 9.13). $\tilde{F}|I \times 0$ lifts $f$ from $e_0$, so is $\tilde{f}$; $\tilde{F}|I \times 1$ lifts $g$ from $\tilde{F}(0,1) = e_0$, so is $\tilde{g}$. As $\tilde{F}(1 \times I)$ is a point, $\tilde{f}(1) = \tilde{g}(1)$. ■

The **lifting correspondence** $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$, $\phi([f]) = \tilde{f}(1)$ with $\tilde{f}$ the lifting from $e_0$, is well defined by Theorem 9.14.

**Theorem 9.15 (54.4)**

If $E$ is path connected, $\phi$ is surjective; if $E$ is simply connected, $\phi$ is bijective.

**Proof.** For $e_1 \in p^{-1}(b_0)$ and a path $\gamma$ in $E$ from $e_0$ to $e_1$, $\gamma$ lifts $f = p \circ \gamma$, so $\phi([f]) = e_1$. If $\phi([f]) = \phi([g])$ and $E$ is simply connected, $\tilde{f} \simeq_p \tilde{g}$ by some $\tilde{F}$ (Lemma 9.6), and $p \circ \tilde{F} : f \simeq_p g$. ■

**Theorem 9.16 (54.5)**

$\pi_1(S^1, b_0)$ is isomorphic to the additive group $\mathbb{Z}$.

**Proof.** For $p : \mathbb{R} \to S^1$, $e_0 = 0$: $p^{-1}(b_0) = \mathbb{Z}$ and $\mathbb{R}$ is simply connected, so $\phi$ is a bijection onto $\mathbb{Z}$. If $\tilde{f}, \tilde{g}$ lift $f, g$ from 0 with $\tilde{f}(1) = n$, $\tilde{g}(1) = m$, then $n + \tilde{g}$ lifts $g$ from $n$ ($p(n + x) = p(x)$), so $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ from 0 to $n + m$: $\phi([f] * [g]) = \phi([f]) + \phi([g])$. ■

So $p_0(s) = p(s)$ generates $\pi_1(S^1, b_0)$, and a loop's class is the end point $n$ of its lifting, the number of times it winds around.

**Theorem 9.17 (54.6)**

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$.
(a) $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective.
(b) Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence induces an injective map $\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$ of the right cosets of $H$, bijective if $E$ is path connected.
(c) A loop $f$ at $b_0$ has $[f] \in H$ iff $f$ lifts to a loop at $e_0$.

**Proof.** (a) If $\tilde{h}$ is a loop at $e_0$ and $F : p \circ \tilde{h} \simeq_p e_{b_0}$, the lifting $\tilde{F}$ from $e_0$ is a path homotopy between $\tilde{h}$ and the constant loop.

(c) If $[f] = [p \circ \tilde{h}]$, $\tilde{h}$ a loop at $e_0$, then $\tilde{f}$ ends where $\tilde{h}$ does, at $e_0$ (Theorem 9.14). Conversely, if $\tilde{f}$ is a loop, $[f] = p_*[\tilde{f}]$.

(b) Let $\Phi(H * [f]) = \phi([f])$. For $[h] \in H$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at $e_0$, $\tilde{h} * \tilde{f}$ lifts $h * f$ and ends at $\tilde{f}(1)$; so $\Phi$ is well defined. If $\phi([f]) = \phi([g])$, $\tilde{f} * \bar{\tilde{g}}$ is a loop at $e_0$ lifting $f * \bar{g}$, so $[f] * [g]^{-1} \in H$, $H * [f] = H * [g]$. Surjectivity: Theorem 9.15. ■

## 9.5 Retractions and fixed points

**Motivation.** $\pi_1(S^1) \neq 0$ forbids retracting the disc onto its boundary, giving Brouwer's theorem in dimension two.

$A \subset X$ is a **retract** of $X$ if there is a continuous $r : X \to A$ with $r|A = \mathrm{id}_A$ (a **retraction**).

**Lemma 9.18 (55.1)**

If $A$ is a retract of $X$, the homomorphism $j_*$ induced by inclusion $j : A \to X$ is injective.

**Proof.** $r \circ j = \mathrm{id}_A$, so $r_* \circ j_* = \mathrm{id}$. ■

**Theorem 9.19 (55.2, no-retraction theorem)**

There is no retraction of $B^2$ onto $S^1$.

**Proof.** $j_* : \pi_1(S^1) \cong \mathbb{Z} \to \pi_1(B^2) = 0$ is not injective. ■

**Lemma 9.20 (55.3)**

For continuous $h : S^1 \to X$ the following are equivalent:
(1) $h$ is nulhomotopic;
(2) $h$ extends to a continuous $k : B^2 \to X$;
(3) $h_*$ is the trivial homomorphism.

**Proof.** (1)⇒(2): let $H : S^1 \times I \to X$ be a homotopy from $h$ to a constant. $\pi(x,t) = (1-t)x$, $S^1 \times I \to B^2$, is continuous, closed and onto, a quotient map collapsing only $S^1 \times 1$ (to 0). $H$ is constant on $S^1 \times 1$, so induces $k$ with $k \circ \pi = H$, extending $h$. (2)⇒(3): $h = k \circ j$, $j : S^1 \to B^2$, and $j_*$ is trivial. (3)⇒(1): for the standard loop $p_0$, $b_0 = p_0(0)$, $h \circ p_0 \simeq_p$ const by some $F$, with $F(0,t) = F(1,t) = F(s,1) = h(b_0)$. The quotient map $q(s,t) = (1-t)p_0(s)$, $I \times I \to B^2$, identifies $(0,t) \sim (1,t)$ and collapses $I \times 1$; $F$ respects this, inducing $k : B^2 \to X$ extending $h$. Then $H(x,t) = k((1-t)x)$ is a homotopy from $h$ to the constant $k(0)$. ■

**Corollary 9.21 (55.4)**

The inclusion $j : S^1 \to \mathbb{R}^2 - 0$ and the identity $i : S^1 \to S^1$ are not nulhomotopic.

**Proof.** $x \mapsto x/\|x\|$ retracts $\mathbb{R}^2 - 0$ onto $S^1$, so $j_*$ is injective (Lemma 9.18), hence nontrivial; $i_*$ is the identity of $\mathbb{Z}$. Apply Lemma 9.20. ■

**Theorem 9.22 (55.5)**

Given a nonvanishing vector field on $B^2$ (a continuous $v : B^2 \to \mathbb{R}^2 - 0$), there is a point of $S^1$ where $v$ points directly inward and one where it points directly outward.

**Proof.** Suppose $v(x) \neq ax$ for all $x \in S^1$, $a < 0$. $w = v|S^1 : S^1 \to \mathbb{R}^2 - 0$ extends to $B^2$, so is nulhomotopic. $F(x,t) = tx + (1-t)w(x)$ never vanishes on $S^1 \times I$ (for $0 < t < 1$, $F = 0$ means $w(x) = -\frac{t}{1-t}x$), so $j \simeq w$ is nulhomotopic, contradicting Corollary 9.21. For outward, use $-v$. ■

**Theorem 9.23 (55.6, Brouwer fixed-point theorem for the disc)**

Every continuous $f : B^2 \to B^2$ has a fixed point.

**Proof.** Otherwise $v(x) = f(x) - x$ is a nonvanishing vector field. At $x \in S^1$, $f(x) - x = ax$ with $a > 0$ would give $f(x) = (1 + a)x \notin B^2$, so $v$ never points directly outward, contradicting Theorem 9.22. ■

**Corollary 9.24 (55.7)**

A $3 \times 3$ matrix $A$ of positive real numbers has a positive real eigenvalue.

**Proof.** $B = \{x \in S^2 : x_1, x_2, x_3 \geq 0\}$ is homeomorphic (by $x \mapsto x/(x_1 + x_2 + x_3)$) to a closed triangle, hence to $B^2$. $Ax$ has positive entries, so $x \mapsto Ax/\|Ax\|$ maps $B$ into $B$; a fixed point gives $Ax = \|Ax\|x$. ■

**Theorem 9.25 (55.8)**

Let $T$ be a closed triangle in $\mathbb{R}^2$. There is $\epsilon > 0$ such that for every open covering $\mathcal{A}$ of $T$ by sets of diameter less than $\epsilon$, some point of $T$ belongs to at least three elements of $\mathcal{A}$.

**Proof.** No point lies on all three edges $e_1, e_2, e_3$, so $(a, b, c) \mapsto \max(|a-b|, |b-c|, |a-c|)$ has a positive minimum $\epsilon$ on the compact $e_1 \times e_2 \times e_3$, and no set of diameter $< \epsilon$ meets all three edges. Suppose no point lies in three elements of such an $\mathcal{A}$; pass to a finite subcovering with a dominated partition of unity $\{\phi_A\}$, and choose a vertex $v_A$ with $A$ missing the opposite edge. Let $k(x) = \sum_A \phi_A(x) v_A$: a convex combination of at most two vertices, so in $\partial T$. If $x \in e$ and $\phi_A(x) > 0$, $A$ meets $e$, so $v_A \in e$; thus $k(e) \subset e$, and $H(x, t) = (1 - t)x + tk(x)$ is a homotopy in $\partial T$ from $\mathrm{id}_{\partial T}$ to $k|\partial T$. But $k|\partial T$ extends to $T$, so is nulhomotopic (Lemma 9.20, $T \cong B^2$), and then so is $\mathrm{id}_{\partial T}$, $\partial T \cong S^1$, contradicting Corollary 9.21. ■

So the triangle has topological dimension at least 2.

## 9.6 The fundamental theorem of algebra

**Motivation.** A degree-$n$ polynomial winds a large circle $n$ times around 0, impossible if it had no root.

**Theorem 9.26 (56.1, fundamental theorem of algebra)**

A polynomial equation $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ of degree $n > 0$ with real or complex coefficients has a complex root.

**Proof.**

**Step 1.** $f(z) = z^n$, $S^1 \to S^1$, induces an injective homomorphism: $f \circ p_0(s) = p(ns)$ lifts to $s \mapsto ns$, so $f_*$ is multiplication by $n$.

**Step 2.** $g(z) = z^n$, $S^1 \to \mathbb{R}^2 - 0$, equals $j \circ f$, so $g_* = j_* f_*$ is injective and $g$ is not nulhomotopic.

**Step 3.** If $\sum |a_i| < 1$, there is a root in $B^2$. Otherwise $k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ maps $B^2$ into $\mathbb{R}^2 - 0$, so $h = k|S^1$ is nulhomotopic. But $F(z, t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on $S^1 \times I$, as $|t(a_{n-1}z^{n-1} + \cdots + a_0)| \leq \sum |a_i| < 1 = |z^n|$; so $g \simeq h$ in $\mathbb{R}^2 - 0$, a contradiction.

**Step 4.** In general choose $c > 0$ with $\sum_i |a_{n-i}|/c^i < 1$; $y^n + (a_{n-1}/c)y^{n-1} + \cdots + a_0/c^n = 0$ has a root $y_0$, and $cy_0$ is a root of the original. ■

## 9.7 The Borsuk–Ulam theorem

**Motivation.** Some point on the earth has the same temperature and pressure as its antipode.

$h : S^n \to S^m$ is **antipode-preserving** if $h(-x) = -h(x)$.

**Theorem 9.27 (57.1)**

A continuous antipode-preserving $h : S^1 \to S^1$ is not nulhomotopic.

**Proof.** After a rotation (commuting with $x \mapsto -x$), $h(b_0) = b_0$. In $S^1 \subset \mathbb{C}$, $q(z) = z^2$ is a quotient map identifying $z$ with $-z$; as $q(h(-z)) = q(h(z))$, there is a continuous $k$ with $k \circ q = q \circ h$. $q_*$ is injective ($q \circ p_0$ lifts to $s \mapsto 2s$). For $f(s) = e^{\pi i s}$, from $b_0$ to $-b_0$, $q \circ f = p_0$; $h \circ f$ runs from $b_0$ to $-b_0$, so its lifting $\tilde{g}$ from 0 ends at some $m + \frac{1}{2}$, and $q \circ h \circ f = p \circ 2\tilde{g}$ with $2\tilde{g}$ ending at $2m + 1$. So $k_*[p_0] = [q \circ h \circ f]$ corresponds to $2m+1 \neq 0$ and $k_*$ is injective. If $h$ were nulhomotopic, $k_* q_* = q_* h_*$ would be trivial, a contradiction. ■

**Theorem 9.28 (57.2)**

There is no continuous antipode-preserving map $g : S^2 \to S^1$.

**Proof.** Its restriction $h$ to the equator is antipode-preserving, hence not nulhomotopic, but extends to the upper hemisphere $\cong B^2$ (via $(x, y, z) \mapsto (x, y)$), so is nulhomotopic (Lemma 9.20). ■

**Theorem 9.29 (57.3, Borsuk–Ulam theorem for $S^2$)**

For every continuous $f : S^2 \to \mathbb{R}^2$ there is $x \in S^2$ with $f(x) = f(-x)$.

**Proof.** Otherwise $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is a continuous antipode-preserving map $S^2 \to S^1$. ■

**Theorem 9.30 (57.4, bisection theorem)**

Given two bounded polygonal regions in $\mathbb{R}^2$, some line bisects each of them.

**Proof.** Put $A_1, A_2$ in the plane $z = 1$ of $\mathbb{R}^3$. For $u \in S^2$ let $f_i(u)$ be the area of the part of $A_i$ with $x \cdot u > 0$; $f_i(-u) = \text{area}(A_i) - f_i(u)$, and $f_i$ is continuous (assumed). Borsuk–Ulam for $(f_1, f_2)$ gives $u$ with $f_i(u) = \frac{1}{2}\,\text{area}(A_i)$. As $u \neq \pm(0, 0, 1)$ (there $f_i(u)$ is 0 or the full area), the plane $x \cdot u = 0$ meets $z = 1$ in a line bisecting both regions. ■

## 9.8 Deformation retracts and homotopy type

**Motivation.** Homotopy equivalence lets us replace a space by a simpler one, e.g. $\mathbb{R}^2 - 0$ by $S^1$.

**Lemma 9.31 (58.1)**

If $h, k : (X, x_0) \to (Y, y_0)$ are homotopic by a homotopy keeping $x_0$ at $y_0$, then $h_* = k_*$.

**Proof.** For a loop $f$ at $x_0$, $G(s, t) = H(f(s), t)$ is a path homotopy from $h \circ f$ to $k \circ f$. ■

**Theorem 9.32 (58.2)**

The inclusion $j : S^n \to \mathbb{R}^{n+1} - 0$ induces an isomorphism of fundamental groups.

**Proof.** $r(x) = x/\|x\|$ has $r \circ j = \mathrm{id}$, and $H(x,t) = (1-t)x + tx/\|x\|$ is a homotopy in $\mathbb{R}^{n+1} - 0$ from the identity to $j \circ r$ fixing $S^n$. By Lemma 9.31, $j_* r_* = \mathrm{id}$, and $r_* j_* = \mathrm{id}$. ■

$A \subset X$ is a **deformation retract** of $X$ if there is $H : X \times I \to X$ with $H(x,0) = x$, $H(x,1) \in A$, $H(a,t) = a$ for $a \in A$.

**Theorem 9.33 (58.3)**

If $A$ is a deformation retract of $X$ and $x_0 \in A$, then $j : (A, x_0) \to (X, x_0)$ induces an isomorphism.

**Proof.** As in Theorem 9.32, with $r(x) = H(x,1)$. ■

**Example 9.4**

$\mathbb{R}^2 - \{p, q\}$ deformation retracts onto a figure eight and onto a theta space; the solid torus $B^2 \times S^1$ and $\mathbb{R}^3$ minus the $z$-axis deformation retract onto a circle, so have $\pi_1 \cong \mathbb{Z}$.

$f : X \to Y$ is a **homotopy equivalence** with **homotopy inverse** $g : Y \to X$ if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$; then $X, Y$ have the same **homotopy type**.

**Lemma 9.34 (58.4)**

Let $h, k : X \to Y$ be homotopic by $H$, with $h(x_0) = y_0$, $k(x_0) = y_1$, and let $\alpha(t) = H(x_0, t)$. Then $k_* = \hat{\alpha} \circ h_*$.

**Proof.** For a loop $f$ at $x_0$ let $G(s,t) = H(f(s), t)$. In the convex $I \times I$, $\beta_0 * \gamma_1 \simeq_p \gamma_0 * \beta_1$, where $\beta_j(s) = (s, j)$, $\gamma_j(t) = (j, t)$; applying $G$, $(h \circ f) * \alpha \simeq_p \alpha * (k \circ f)$, so $[k \circ f] = [\bar{\alpha}] * [h \circ f] * [\alpha]$. ■

**Corollary 9.35 (58.5)**

If $h \simeq k$ and $h_*$ is injective, surjective or trivial, so is $k_*$.

**Corollary 9.36 (58.6)**

If $h$ is nulhomotopic, $h_*$ is trivial.

**Proof.** Both follow from Lemma 9.34: $\hat{\alpha}$ is an isomorphism, and constant maps induce the trivial homomorphism. ■

**Theorem 9.37 (58.7)**

If $f : X \to Y$ is a homotopy equivalence with $f(x_0) = y_0$, then $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ is an isomorphism.

**Proof.** Let $g$ be a homotopy inverse, $x_1 = g(y_0)$, $y_1 = f(x_1)$. By Lemma 9.34, $g \circ f \simeq \mathrm{id}_X$ makes $g_* \circ (f_{x_0})_* : \pi_1(X, x_0) \to \pi_1(X, x_1)$ an isomorphism, so $(g_{y_0})_*$ is onto; $f \circ g \simeq \mathrm{id}_Y$ makes $(f_{x_1})_* \circ (g_{y_0})_*$ an isomorphism, so $(g_{y_0})_*$ is injective. Hence $(g_{y_0})_*$, and so $(f_{x_0})_*$, is an isomorphism. ■

## 9.9 The fundamental group of $S^n$

**Motivation.** Higher spheres are simply connected, by computing $\pi_1$ from an open cover (made systematic in Chapter 11).

**Theorem 9.38 (59.1)**

Let $X = U \cup V$ with $U, V$ open, $U \cap V$ path connected and $x_0 \in U \cap V$. Then the images of $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and $j_* : \pi_1(V, x_0) \to \pi_1(X, x_0)$ generate $\pi_1(X, x_0)$.

**Proof.** For a loop $f$ at $x_0$ choose (Lebesgue) $0 = a_0 < \cdots < a_n = 1$ with each $f([a_{i-1}, a_i])$ in $U$ or $V$. If some $f(a_i)$, $0 < i < n$, lies in $U - V$, say, both adjacent intervals map into $U$ and $a_i$ can be deleted; so assume all $f(a_i) \in U \cap V$. Choose paths $\alpha_i$ in $U \cap V$ from $x_0$ to $f(a_i)$ ($\alpha_0, \alpha_n$ constant). With $f_i$ as in Theorem 9.3, $[f] = [f_1] * \cdots * [f_n] = [g_1] * \cdots * [g_n]$, where $g_i = \alpha_{i-1} * f_i * \bar{\alpha}_i$ is a loop at $x_0$ in $U$ or $V$. ∎

**Corollary 9.39 (59.2)**

If $X = U \cup V$ with $U, V$ open and simply connected and $U \cap V$ nonempty and path connected, then $X$ is simply connected.

**Theorem 9.40 (59.3)**

For $n \geq 2$, $S^n$ is simply connected.

**Proof.** Let $p = (0, \ldots, 0, 1)$. Stereographic projection $f(x) = (x_1, \ldots, x_n)/(1 - x_{n+1})$ is a homeomorphism $S^n - p \to \mathbb{R}^n$ with inverse $y \mapsto (t(y)y, 1 - t(y))$, $t(y) = 2/(1 + \|y\|^2)$; similarly $S^n - (-p) \cong \mathbb{R}^n$. Both are simply connected, and their intersection is $\cong \mathbb{R}^n - 0$, path connected for $n \geq 2$. Apply Corollary 9.39. ∎

## 9.10 Fundamental groups of some surfaces

**Motivation.** $\pi_1$ distinguishes the sphere, torus, projective plane and double torus.

A **surface** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^2$.

**Theorem 9.41 (60.1)**

$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.

**Proof.** With projections $p, q$, $\Phi([f]) = (p_*[f], q_*[f])$ is a homomorphism. For loops $g$ in $X$, $h$ in $Y$, $f(s) = g(s) \times h(s)$ maps to $([g], [h])$. If $G : p \circ f \simeq_p e$ and $H : q \circ f \simeq_p e$, then $G \times H : f \simeq_p e$. ∎

**Corollary 9.42 (60.2)**

The fundamental group of the torus $S^1 \times S^1$ is $\mathbb{Z} \times \mathbb{Z}$.

The **projective plane** $P^2$ is the quotient of $S^2$ obtained by identifying each $x$ with $-x$.

**Theorem 9.43 (60.3)**

$P^2$ is a compact surface, and the quotient map $p : S^2 \to P^2$ is a covering map.

**Proof.** $p$ is open: $p^{-1}(p(U)) = U \cup a(U)$, $a(x) = -x$. For $x \in S^2$ the $\epsilon$-neighbourhood $U$ ($\epsilon < 1$) contains no antipodal pair, so $p|U$ and $p|a(U)$ are continuous open bijections onto $p(U)$, homeomorphisms, and $a(U) \cap U = \emptyset$: $p(U)$ is evenly covered. Hausdorff: for $p(x) \neq p(y)$, $\pm x, \pm y$ are distinct; for $\epsilon$ less than half their minimum distance, the $\epsilon$-neighbourhoods $U \ni x$, $W \ni y$ have $(U \cup a(U)) \cap (W \cup a(W)) = \emptyset$, so $p(U) \cap p(W) = \emptyset$. $p$ open carries a countable basis of $S^2$ to one of $P^2$; $P^2$ is locally homeomorphic to $S^2$, hence to $\mathbb{R}^2$, and compact as an image of $S^2$. ■

**Corollary 9.44 (60.4)**

$\pi_1(P^2, y)$ has order 2.

**Proof.** $S^2$ is simply connected, so the lifting correspondence is a bijection onto the two-point fibre (Theorem 9.15). ■

Similarly $\pi_1(P^n)$ ($n \geq 2$) has order 2.

**Lemma 9.45 (60.5)**

The fundamental group of the figure eight is not abelian.

**Proof.** Let $X = A \cup B$, circles meeting at $x_0$. Let $E \subset \mathbb{R}^2$ be the union of the coordinate axes and, for each integer $n \neq 0$, a circle tangent to the $x$-axis at $(n, 0)$ and one tangent to the $y$-axis at $(0, n)$. Let $p : E \to X$ wrap the $x$-axis around $A$ (each $[n, n+1]$ once, integers to $x_0$) and the $y$-axis around $B$, and map each circle tangent to the $x$-axis onto $B$, each tangent to the $y$-axis onto $A$ (tangency points to $x_0$). Small arcs in $X - x_0$ and a small "cross" at $x_0$ have preimages that are disjoint arcs, resp. crosses about the lattice points on the axes, mapped homeomorphically; so $p$ is a covering map. For loops $g$, $h$ once around $A$, $B$, the lifting of $g * h$ from the origin runs to $(1, 0)$ and around the circle tangent there, ending at $(1, 0)$; that of $h * g$ ends at $(0, 1)$. By Theorem 9.14, $g * h \not\simeq_p h * g$. ■

**Theorem 9.46 (60.6)**

The fundamental group of the double torus is not abelian.

**Proof.** The double torus $X$ contains a figure eight $A$ (a circle around each handle, meeting at a point), a retract of $X$: squash each handle onto its circle, compatibly along the joining curve. By Lemma 9.18, $\pi_1(X)$ contains the nonabelian $\pi_1(A)$. ■

**Corollary 9.47 (60.7)**

The 2-sphere, torus, projective plane and double torus are topologically distinct.

**Proof.** Their fundamental groups are trivial, $\mathbb{Z} \times \mathbb{Z}$ (infinite abelian), of order 2, and nonabelian. ■

# Chapter 10. Separation Theorems in the Plane

## 10.1 The Jordan separation theorem

**Motivation.** Fundamental groups show that a simple closed curve disconnects the plane; we start with this easier half.

Open subsets of $S^2$ and $\mathbb{R}^2$ are locally path connected, so their components are open and equal their path components. $A$ **separates** a connected $X$ if $X - A$ is not connected (into $n$ components if $X - A$ has $n$ components). An **arc** is a space homeomorphic to $[0,1]$; a **simple closed curve** one homeomorphic to $S^1$.

**Lemma 10.1 (61.1)**

Let $C \subset S^2$ be compact, $b \in S^2 - C$, and $h : S^2 - b \to \mathbb{R}^2$ a homeomorphism. If $U$ is a component of $S^2 - C$ not containing $b$, then $h(U)$ is a bounded component of $\mathbb{R}^2 - h(C)$; if $U$ contains $b$, then $h(U - b)$ is the unbounded component. In particular $S^2 - C$ and $\mathbb{R}^2 - h(C)$ have the same number of components.

**Proof.** $U - b$ is connected for every component $U$: if $b \in U$ and $A, B$ separate $U - b$, take a disc neighbourhood $W \subset U$ of $b$ missing $C$; the connected $W - b$ lies in $A$, say, so $b$ is not a limit point of $B$ and $A \cup \{b\}$, $B$ separate $U$. Hence the $h(U_\alpha - b)$, $U_\alpha$ the components of $S^2 - C$, are disjoint connected open sets with union $\mathbb{R}^2 - h(C)$, its components. Extend $h$ to $H : S^2 \to \mathbb{R}^2 \cup \{\infty\}$, $H(b) = \infty$. For $U \ni b$, $H(U)$ is a neighbourhood of $\infty$, so $h(U - b)$ contains the complement of a compact set: it is unbounded, and the other components are bounded. ■

**Lemma 10.2 (61.2, nulhomotopy lemma)**

Let $a, b \in S^2$, $A$ compact and $f : A \to S^2 - a - b$ continuous. If $a$ and $b$ lie in the same component of $S^2 - f(A)$, then $f$ is nulhomotopic.

**Proof.** With $S^2 = \mathbb{R}^2 \cup \{\infty\}$, $a = 0$, $b = \infty$: if $g : A \to \mathbb{R}^2 - 0$ and 0 is in the unbounded component of $\mathbb{R}^2 - g(A)$, then $g$ is nulhomotopic. Take a ball $B \ni 0$ containing $g(A)$, $p \notin B$, and a path $\alpha$ in $\mathbb{R}^2 - g(A)$ from 0 to $p$. $G(x,t) = g(x) - \alpha(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from $g$ to $k(x) = g(x) - p$, and $H(x,t) = tg(x) - p$ one from the constant $-p$ to $k$ ($tg(x) \in B$, $p \notin B$). ■

**Theorem 10.3 (61.3, Jordan separation theorem)**

A simple closed curve $C$ in $S^2$ separates $S^2$.

**Proof.** Suppose $S^2 - C$ is connected, hence path connected. Write $C = A_1 \cup A_2$, arcs meeting in their end points $a, b$; $X = S^2 - a - b$, $U = S^2 - A_1$, $V = S^2 - A_2$. Then $X = U \cup V$, $U \cap V = S^2 - C$ path connected. If $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and similarly $j_*$ are trivial, $\pi_1(X, x_0) = 1$ (Theorem 9.38), whereas $X \cong \mathbb{R}^2 - 0$. A loop $f$ in $U$ at $x_0$ induces $h : S^1 \to U$, $h \circ p_0 = f$. $h(S^1)$ misses the connected $A_1 \ni a, b$, so $i \circ h$ is nulhomotopic (Lemma 10.2), $(i \circ h)_*$ is trivial (Lemma 9.20), and $i_*[f] = (i \circ h)_*[p_0]$. ■

**Theorem 10.4 (61.4, general separation theorem)**

If $A_1, A_2$ are closed connected subsets of $S^2$ with $A_1 \cap A_2 = \{a, b\}$, then $C = A_1 \cup A_2$ separates $S^2$.

**Proof.** $C \neq S^2$: $S^2 - a - b$ is connected, $C - a - b$ is separated by the $A_i - a - b$. The proof of Theorem 10.3 used only that $A_1, A_2$ are closed, connected and meet in $\{a, b\}$. ■

## 10.2 Invariance of domain

**Motivation.** A continuous injection of an open $U \subseteq \mathbb{R}^2$ into $\mathbb{R}^2$ has open image.

**Lemma 10.5 (62.1, homotopy extension lemma)**

Let $X \times I$ be normal, $A \subset X$ closed, and $f : A \to Y$ continuous with $Y$ open in $\mathbb{R}^n$. If $f$ is nulhomotopic, it extends to a continuous nulhomotopic $g : X \to Y$.

**Proof.** Let $F : A \times I \to Y$ be a homotopy from $f$ to the constant $y_0$; set $F(x, 1) = y_0$ on $X$. $F$ is continuous on the closed $(A \times I) \cup (X \times 1)$, so extends (Tietze, coordinatewise) to $G : X \times I \to \mathbb{R}^n$. $U = G^{-1}(Y)$ is open and contains $(A \times I) \cup (X \times 1)$; by the tube lemma there is an open $W \supset A$ with $W \times I \subset U$. Urysohn ($X \cong X \times 0$ normal) gives $\phi : X \to [0, 1]$, $\phi = 0$ on $A$, $\phi = 1$ off $W$. Then $(x, \phi(x)) \in U$, so $g(x) = G(x, \phi(x))$ maps $X$ into $Y$ and extends $f$, and $H(x, t) = G(x, (1-t)\phi(x)+t)$ stays in $Y$ and joins $g$ to $y_0$. ■

**Lemma 10.6 (62.2, Borsuk lemma)**

Let $a, b \in S^2$, $A$ compact, and $f : A \to S^2 - a - b$ continuous and injective. If $f$ is nulhomotopic, then $a$ and $b$ lie in the same component of $S^2 - f(A)$.

**Proof.** $f(A) \cong A$, so take $f$ an inclusion, $a = 0$, $b = \infty$: if $A \subset \mathbb{R}^2 - 0$ is compact and $j : A \to \mathbb{R}^2 - 0$ is nulhomotopic, 0 is in the unbounded component of $\mathbb{R}^2 - A$. Suppose the component $C \ni 0$ is bounded, $D$ the union of the others (both open). $C \cup A = \mathbb{R}^2 - D$ is compact, and Lemma 10.5 extends $j$ to $k : C \cup A \to \mathbb{R}^2 - 0$. Let $h = k$ on $C \cup A$, $h(x) = x$ on $D \cup A$; $h : \mathbb{R}^2 \to \mathbb{R}^2 - 0$ is continuous (pasting lemma). For a closed ball $B$ of radius $M$ with $C \cup A \subset \operatorname{Int} B$, $h$ is the identity on $\operatorname{Bd} B$, so $x \mapsto Mh(x)/\|h(x)\|$ retracts $B$ onto $\operatorname{Bd} B$, contradicting Theorem 9.19. ■

**Theorem 10.7 (62.3, invariance of domain)**

If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open and $f^{-1} : f(U) \to U$ is continuous.

**Proof.** It suffices that $f$ is an open map into $S^2 \supset \mathbb{R}^2$.

**Step 1.** For a closed ball $B \subset U$, $f(B)$ does not separate $S^2$: for $a, b \notin f(B)$, $f|B : B \to S^2 - a - b$ is nulhomotopic ($B$ contractible) and injective; apply Lemma 10.6.

**Step 2.** $f(\operatorname{Int} B)$ is open. $C = f(\operatorname{Bd} B)$ separates $S^2$ (Theorem 10.3). Let $V$ be the component of $S^2 - C$ containing the connected $f(\operatorname{Int} B)$, $W \neq \emptyset$ the union of the others. If $a \in V - f(\operatorname{Int} B)$, then $a \notin f(B)$; for $b \in W$, Step 1 puts $a, b$ in a connected subset of $S^2 - f(B) \subset S^2 - C$, a contradiction. So $f(\operatorname{Int} B) = V$.

**Step 3.** Open subsets of $U$ are unions of interiors of closed balls in $U$, so $f$ is open. ■

The theorem holds in $\mathbb{R}^n$, but the proof needs higher homotopy or homology groups.

## 10.3 The Jordan curve theorem

**Motivation.** The complement of a simple closed curve has exactly two components, each with boundary the curve.

**Theorem 10.8 (63.1)**

Let $X = U \cup V$ with $U, V$ open and $U \cap V = A \cup B$, $A, B$ disjoint and open. Let $\alpha$ be a path in $U$ from $a \in A$ to $b \in B$, $\beta$ a path in $V$ from $b$ to $a$, and $f = \alpha * \beta$.
(a) $[f]$ generates an infinite cyclic subgroup of $\pi_1(X, a)$.
(b) If $\pi_1(X, a)$ is infinite cyclic, it is generated by $[f]$.
(c) If $\gamma$ is a path in $U$ from $a$ to $a' \in A$ and $\delta$ a path in $V$ from $a'$ to $a$, and $g = \gamma * \delta$, then the subgroups generated by $[f]$ and $[g]$ meet only in the identity.

**Proof.**

**Step 1 (a covering space).** Let $Y \subset X \times \mathbb{Z}$ be the disjoint union of $U \times 2n$ and $V \times (2n+1)$, $n \in \mathbb{Z}$, and $E$ its quotient identifying $x \times 2n \sim x \times (2n+1)$ for $x \in B$ and $x \times 2n \sim x \times (2n-1)$ for $x \in A$; $\pi : Y \to E$ the quotient map, and $p : E \to X$ induced by $x \times m \mapsto x$. $\pi$ is open: for $W$ open in $U$, $\pi^{-1}\pi(W \times 2n) = W \times 2n \cup (W \cap B) \times (2n+1) \cup (W \cap A) \times (2n-1)$; similarly for $V$. So $p^{-1}(U)$ is the disjoint union of the open $\pi(U \times 2n)$, each mapped homeomorphically onto $U$ ($\pi | U \times 2n$ a continuous open bijection); likewise for $V$. So $p$ is a covering map.

**Step 2.** Let $e_n = \pi(a \times 2n)$, the distinct points of $p^{-1}(a)$. $\tilde{\alpha}_n(s) = \pi(\alpha(s) \times 2n)$ and $\tilde{\beta}_n(s) = \pi(\beta(s) \times (2n+1))$ lift $\alpha, \beta$; $\tilde{\alpha}_n$ ends at $\pi(b \times 2n) = \pi(b \times (2n+1))$, where $\tilde{\beta}_n$ begins, and $\tilde{\beta}_n$ ends at $\pi(a \times (2n+1)) = e_{n+1}$. So $\tilde{f}_n = \tilde{\alpha}_n * \tilde{\beta}_n$ lifts $f$ from $e_n$ to $e_{n+1}$.

**Step 3 (a).** For $m > 0$, $\tilde{f}_0 * \cdots * \tilde{f}_{m-1}$ lifts $f^m$ from $e_0$ to $e_m \neq e_0$, so $[f]^m \neq 1$.

**Step 4 (b).** The lifting correspondence $\phi$ sends $[f]^m \mapsto e_m$ ($m \in \mathbb{Z}$), so is onto. By Theorem 9.17 it induces a bijection $\pi_1(X, a)/H \to p^{-1}(a)$, $H = p_*\pi_1(E, e_0)$; the fibre is infinite while $\mathbb{Z}$ modulo a nontrivial subgroup is finite, so $H = 1$ and $\phi$ is bijective. As $\phi$ maps $\langle [f] \rangle$ onto the fibre, $\langle [f] \rangle = \pi_1(X, a)$.

**Step 5 (c).** $\tilde{\gamma}(s) = \pi(\gamma(s) \times 0)$ and $\tilde{\delta}(s) = \pi(\delta(s) \times (-1))$ lift $\gamma, \delta$, with $\pi(a' \times 0) = \pi(a' \times (-1))$ ($a' \in A$) and $\tilde{\delta}(1) = \pi(a \times (-1)) = e_0$. So every power of $g$ lifts to a loop at $e_0$, while $f^m$ ($m \neq 0$) lifts to a path ending at $e_m \neq e_0$. Hence $[f]^m = [g]^k$ forces $m = 0$ and $[g]^k = 1$. ∎

**Theorem 10.9 (63.2, nonseparation theorem)**

An arc $D$ in $S^2$ does not separate $S^2$.

**Proof.** $D$ is contractible, so for $a, b \notin D$ the inclusion $D \to S^2 - a - b$ is nulhomotopic; apply Lemma 10.6. ∎

A second proof gives a generalization.

**Theorem 10.10 (63.3, general nonseparation theorem)**

Let $D_1, D_2$ be closed subsets of $S^2$ with $S^2 - (D_1 \cap D_2)$ simply connected. If neither $D_1$ nor $D_2$ separates $S^2$, neither does $D_1 \cup D_2$.

**Proof.** Let $a, b \notin D_1 \cup D_2$ not be joined by a path in $S^2 - (D_1 \cup D_2)$. Put $X = S^2 - (D_1 \cap D_2)$, $U = S^2 - D_1$, $V = S^2 - D_2$; $X = U \cup V$, $U \cap V = S^2 - (D_1 \cup D_2)$. Let $A$ be the path component of $U \cap V$ containing $a$, $B$ the union of the others (both open). As $D_1, D_2$ do not separate, there are

paths in $U$ from $a$ to $b$ and in $V$ from $b$ to $a$, so $\pi_1(X,a) \neq 1$ by Theorem 10.8(a), a contradiction. ■

(For an arc $D = h([0,1])$ this reproves Theorem 10.9: if $a, b$ cannot be joined in $S^2 - D$, then not in $S^2 - h(I_1)$ for a half $I_1$ of $[0,1]$, as $S^2-$point is simply connected; bisecting gives nested $I_n$ shrinking to $x$, but a path from $a$ to $b$ in $S^2 - h(x)$ misses $h(I_n)$ for large $n$.)

**Theorem 10.11 (63.4, Jordan curve theorem)**

A simple closed curve $C$ in $S^2$ separates $S^2$ into precisely two components $W_1, W_2$, and $C = \overline{W_i} - W_i$ for $i = 1, 2$.

**Proof.**

**Step 1.** Write $C = C_1 \cup C_2$, arcs meeting in $\{p, q\}$; $X = S^2 - p - q$, $U = S^2 - C_1$, $V = S^2 - C_2$, so $U \cap V = S^2 - C$ has at least two components (Theorem 10.3). Suppose more: $A_1, A_2$ two components, $B$ the union of the others, all open; $a \in A_1$, $a' \in A_2$, $b \in B$. The $C_i$ do not separate (Theorem 10.9), giving paths $\alpha, \gamma$ in $U$ from $a$ to $b$, $a'$ and $\beta, \delta$ in $V$ from $b$, $a'$ to $a$. For $f = \alpha * \beta$, $g = \gamma * \delta$, Theorem 10.8(a) for $(A_1 \cup A_2) \cup B$ and for $A_1 \cup (A_2 \cup B)$ shows $[f], [g]$ have infinite order. As $\pi_1(X,a) \cong \mathbb{Z}$, $[f]^m = [g]^k$ for some nonzero $m, k$, contradicting Theorem 10.8(c) for $(A_1 \cup A_2) \cup B$, where $a, a' \in A_1 \cup A_2$.

**Step 2.** $W_1, W_2$ are open, so $\overline{W_i} - W_i \subset C$. Conversely let $x \in C$, $U \ni x$ a neighbourhood; write $C = C_1 \cup C_2$, arcs meeting in their end points, with $C_1 \subset U$. For $a \in W_1$, $b \in W_2$ take a path $\alpha$ from $a$ to $b$ in $S^2 - C_2$ (Theorem 10.9). Its connected image meets the disjoint open $W_1$ and $S^2 - \overline{W_1} \supset W_2$, so contains a point $y \in \overline{W_1} - W_1 \subset C$, and $y \in C_1 \subset U$. So $x \in \overline{W_1} - W_1$; likewise for $W_2$. ■

**Theorem 10.12 (63.5)**

If $C_1, C_2$ are closed connected subsets of $S^2$ meeting in exactly two points, and neither separates $S^2$, then $C_1 \cup C_2$ separates $S^2$ into precisely two components.

**Proof.** At least two by Theorem 10.4; at most two by Step 1 above, which used only these properties. ■

Two arcs meeting in two points not both end points separate $S^2$ into two components, but need not be their common boundary.

## 10.4 Imbedding graphs in the plane

**Motivation.** The Jordan curve theorem shows $K_{3,3}$ (three houses, three utilities) is not planar.

A **linear graph** is a Hausdorff space that is a finite union of arcs (**edges**), any two meeting in at most a common end point (a **vertex**). The **complete graph on $n$ vertices** has an edge joining each pair of vertices; the **utilities graph** has vertices $g, w, e, h_1, h_2, h_3$ with an edge joining each of $g, w, e$ to each $h_i$. A **theta space** is a Hausdorff space that is the union of three arcs $A, B, C$, each pair meeting precisely in their common end points.

**Lemma 10.13 (64.1)**

A theta space $X = A \cup B \cup C$ in $S^2$ separates $S^2$ into three components, with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The component with boundary $A \cup B$ is a component of $S^2 - (A \cup B)$.

**Proof.** Let $a, b$ be the common end points. $A \cup B$ separates $S^2$ into $U, U'$ with boundary $A \cup B$ (Jordan curve theorem), and the connected $C - a - b$ lies in one, say $U'$. The connected $\overline{U} = U \cup A \cup B$ and $C$ meet only in $\{a, b\}$ and neither separates ($C$ is an arc, $S^2 - \overline{U} = U'$). By Theorem 10.12, $\overline{U} \cup C$ separates $S^2$ into $V, W$. So $S^2 - X = U \cup V \cup W$, disjoint open connected sets, its components; $U$ has boundary $A \cup B$ and is a component of $S^2 - (A \cup B)$. By symmetry the others have boundaries $B \cup C$, $A \cup C$. ■

**Theorem 10.14 (64.2)**

The utilities graph cannot be imbedded in the plane.

**Proof.** Suppose it lies in $S^2$. $A = gh_1 \cup h_1 w$, $B = gh_2 \cup h_2 w$, $C = gh_3 \cup h_3 w$ form a theta space, separating $S^2$ into $U, V, W$ with boundaries $A \cup B$, $B \cup C$, $A \cup C$. $e$ and the connected $eh_i - h_i$ lie in one of them, so $h_1, h_2, h_3$ lie in its boundary. But $A \cup B \not\ni h_3$, $B \cup C \not\ni h_1$, $A \cup C \not\ni h_2$. ■

**Lemma 10.15 (64.3)**

A complete graph $X$ on four vertices $a_1, \ldots, a_4$ in $S^2$ separates $S^2$ into four components, whose boundaries are the subgraphs $X_i$ consisting of the edges not having $a_i$ as a vertex.

**Proof.** The union $Y$ of all edges but $a_2 a_4$ is the theta space with arcs $A = a_1 a_2 \cup a_2 a_3$, $B = a_1 a_3$, $C = a_1 a_4 \cup a_4 a_3$, separating $S^2$ into $U, V, W$ with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The connected $a_2 a_4 - a_2 - a_4$ lies in one whose boundary contains $a_2, a_4$: not $U$ ($a_4 \notin A \cup B$), not $V$ ($a_2 \notin B \cup C$), so $W$. $\overline{U} \cup \overline{V}$ is connected (both contain $B$) and does not separate (complement $W$), the arc $a_2 a_4$ does not separate, and they meet only in $a_2, a_4$; by Theorem 10.12 their union separates $S^2$ into $W_1, W_2$. So $S^2 - X = U \cup V \cup W_1 \cup W_2$, its components. $U$ has boundary $A \cup B = X_4$; by symmetry the others have $X_1, X_2, X_3$. ■

**Theorem 10.16 (64.4)**

The complete graph on five vertices cannot be imbedded in the plane.

**Proof.** Suppose $G \subset S^2$ has vertices $a_1, \ldots, a_5$; let $X$ be the complete subgraph on $a_1, \ldots, a_4$. $a_5$ and the connected set of edges at $a_5$ minus $a_1, \ldots, a_4$ lie in one component of $S^2 - X$, so $a_1, \ldots, a_4$ lie in its boundary $X_i$; but $a_i \notin X_i$. ■

By Kuratowski's theorem, a graph imbeds in the plane iff it contains no subgraph homeomorphic to the utilities graph or the complete graph on five vertices.

## 10.5 Winding number of simple closed curves

**Motivation.** The winding number of a simple closed curve is $\pm 1$ inside and 0 outside.

**Lemma 10.17 (65.1)**

Let $G \subset S^2$ be a complete graph on $a_1, \ldots, a_4$, $C$ the simple closed curve $a_1 a_2 a_3 a_4 a_1$, and $p, q$ interior points of the edges $a_1 a_3$ and $a_2 a_4$. Then (a) $p$ and $q$ lie in different components of $S^2 - C$;
(b) the inclusion $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

**Proof.** (a) The theta space $C \cup a_1a_3$ (arcs $a_1a_2a_3$, $a_1a_3$, $a_1a_4a_3$) separates $S^2$ into three components; one, $W$, has boundary $C$, the only one whose boundary contains $a_2$ and $a_4$. So $a_2a_4 - a_2 - a_4 \subset W$ and $q \in W$. By Lemma 10.13, $W$ is a component of $S^2 - C$, and $p \notin W$ ($p$ is on the theta space).

(b) Let $X = S^2 - p - q$, $x$ interior to $a_1a_2$, $y$ interior to $a_3a_4$, $\alpha$ the path in $C$ from $x$ through $a_2, a_3$ to $y$, $\beta$ the path in $C$ from $y$ through $a_4, a_1$ to $x$. Let $D_1$, $D_2$ be the arcs of $D = a_1a_3a_2a_4a_1$ from $p$ through $a_1, a_4$, resp. $a_3, a_2$, to $q$; $U = S^2 - D_1$, $V = S^2 - D_2$. Then $X = U \cup V$, $U \cap V = S^2 - D$ has two components (Jordan curve theorem), containing $x$ and $y$ respectively by (a) for $G$ with the curve $D$ ("diagonals" $a_1a_2$, $a_3a_4$); $\alpha \subset U$, $\beta \subset V$. As $\pi_1(X, x) \cong \mathbb{Z}$, $[\alpha * \beta]$ generates it (Theorem 10.8(b)). Since $\alpha * \beta$ is a loop in $C$, $j_* : \pi_1(C, x) \cong \mathbb{Z} \to \pi_1(X, x) \cong \mathbb{Z}$ is onto, hence an isomorphism. ■

**Theorem 10.18 (65.2)**

If $C$ is a simple closed curve in $S^2$ and $p, q$ lie in different components of $S^2 - C$, then $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

**Proof.**

**Step 1.** If $A$ is an arc from $a$ to $b$ and $B$ one from $b$ to $c$ ($a, b, c$ distinct), $A \cup B$ contains an arc from $a$ to $c$: for homeomorphisms $f : I \to A$, $g : I \to B$ (from $a$ to $b$, $b$ to $c$), let $t_0$ be least with $f(t_0) \in B$, $g(t_1) = f(t_0)$; take $f([0, t_0]) \cup g([t_1, 1])$.

**Step 2.** Any two points of a connected open $U \subset \mathbb{R}^2$ are end points of an arc in $U$: "equal, or joined by an arc in $U$" is an equivalence relation (Step 1) with open classes (segments in a disc).

**Step 3.** A complete graph on four vertices containing $C$ as $a_1a_2a_3a_4a_1$. In $\mathbb{R}^2$, with 0 in the bounded component of $\mathbb{R}^2 - C$, let $a_1$ be the largest point of $C$ on the negative $x$-axis, $a_3$ the smallest on the positive one; the open segment between them misses $C$ and contains 0, so lies in the bounded component. Write $C = C_1 \cup C_2$, arcs from $a_1$ to $a_3$, and take $a$ in the unbounded component. As $C_1, C_2$ do not separate, Step 2 gives injective paths $\alpha$ in $\mathbb{R}^2 - C_1$, $\beta$ in $\mathbb{R}^2 - C_2$ from $a$ to 0. Let $t_0$ be least with $\alpha(t_0) \in C$, $a_2 = \alpha(t_0)$, interior to $C_2$; similarly $a_4 = \beta(t_1)$ interior to $C_1$. $\alpha([0, t_0])$, $\beta([0, t_1])$ meet $C$ only at their end points and otherwise lie in the unbounded component; by Step 1 their union contains an arc from $a_2$ to $a_4$ meeting $C \cup a_1a_3$ only in $a_2, a_4$. With $a_1a_3$ and $C$ it forms the graph, so by Lemma 10.17 the theorem holds for some $p, q$ in different components.

**Step 4.** Moving $p$ (and symmetrically $q$) in its component. With $q = \infty$: let $D$ be a simple closed curve, $0, p$ in one component of $\mathbb{R}^2 - D$, and $j : D \to \mathbb{R}^2 - 0$ inducing an isomorphism. For $f(x) = x - p$, a homeomorphism $\mathbb{R}^2 - p \to \mathbb{R}^2 - 0$, and a path $\gamma$ in $\mathbb{R}^2 - D$ from 0 to $p$, $F(x, t) = x - \gamma(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from $j$ to $f \circ k$, $k : D \to \mathbb{R}^2 - p$ the inclusion. By Corollary 9.35, $(f \circ k)_*$ and hence $k_*$ are isomorphisms. ■

## 10.6 The Cauchy integral formula

**Motivation.** The facts above justify the Cauchy integral formula for simple closed curves.

For a loop $f$ in $\mathbb{R}^2$ and $a \notin f(I)$, let $g(s) = (f(s) - a)/\|f(s) - a\|$, a loop in $S^1$, and $\tilde{g}$ a lifting to $\mathbb{R}$. The integer $n(f, a) = \tilde{g}(1) - \tilde{g}(0)$ is the **winding number** of $f$ about $a$, independent of the lifting (two differ by a constant). A **free homotopy** of loops is a map $F : I \times I \to X$ with $F(0, t) = F(1, t)$ for all $t$.

**Lemma 10.19 (66.1)**

Let $f$ be a loop in $\mathbb{R}^2 - a$.
(a) $n(\bar{f}, a) = -n(f, a)$.

(b) If $f$ is freely homotopic to $f'$ through loops in $\mathbb{R}^2 - a$, then $n(f,a) = n(f',a)$.
(c) If $a, b$ lie in the same component of $\mathbb{R}^2 - f(I)$, then $n(f,a) = n(f,b)$.

**Proof.** (a) $s \mapsto 1-s$ reverses the sign of $\tilde{g}(1) - \tilde{g}(0)$.

(b) For a free homotopy $F$, lift $G = (F-a)/\|F-a\|$ to $\tilde{G} : I \times I \to \mathbb{R}$ (Lemma 9.13); $\tilde{G}(1,t) - \tilde{G}(0,t)$ is continuous and integer valued, hence constant.

(c) $n(f,a) = n(f-a, 0)$, and for a path $\gamma$ in $\mathbb{R}^2 - f(I)$ from $a$ to $b$, $f(s) - \gamma(t)$ is a free homotopy in $\mathbb{R}^2 - 0$ from $f-a$ to $f-b$. ■

A loop $f$ is **simple** if $f(s) = f(s')$ only when $s = s'$ or $\{s, s'\} = \{0, 1\}$; its image is a simple closed curve.

**Theorem 10.20 (66.2)**

Let $f$ be a simple loop in $\mathbb{R}^2$. If $a$ is in the unbounded component of $\mathbb{R}^2 - f(I)$, $n(f,a) = 0$; if $a$ is in the bounded component, $n(f,a) = \pm 1$.

**Proof.** Take $a = 0$ and, rotating (a free homotopy), the base point $x_0$ on the positive $x$-axis. Let $C = f(I)$, $h : S^1 \to C$ with $h \circ p_0 = f$; $h_*[p_0]$ generates $\pi_1(C, x_0)$. If 0 is in the bounded component, $[f] = j_* h_* [p_0]$ generates $\pi_1(\mathbb{R}^2 - 0, x_0)$ (Theorem 10.18, $\infty$ in the other component); if in the unbounded one, $j \circ h$ is nulhomotopic (Lemma 10.2) and $[f] = 1$. The retraction $x \mapsto x/\|x\|$ induces an isomorphism onto $\pi_1(S^1, b_0)$, so $g = f/\|f\|$ represents a generator or the identity: its lifting from 0 ends at $\pm 1$ or 0 (proof of Theorem 9.16). ■

A simple loop is **counterclockwise** if $n(f,a) = +1$ for $a$ in the bounded component and **clockwise** if it is $-1$.

**Lemma 10.21 (66.3)**

If $f$ is a piecewise-differentiable loop in $\mathbb{C}$ and $a \notin f(I)$, then $n(f,a) = \dfrac{1}{2\pi i} \displaystyle\int_f \frac{dz}{z-a}$.

**Proof.** Let $r = |f-a|$, $g = (f-a)/r$, $\tilde{g}$ a lifting, $\theta = 2\pi\tilde{g}$; then $f(s) - a = r(s)e^{i\theta(s)}$, $r, \theta$ piecewise differentiable ($\theta$ locally a branch of the argument), and

$$\int_f \frac{dz}{z-a} = \int_0^1 \frac{f'(s)}{f(s)-a}\,ds = \int_0^1 \Big(\frac{r'(s)}{r(s)} + i\theta'(s)\Big)ds = [\log r]_0^1 + i[\theta]_0^1 = 2\pi i\, n(f,a).$$

■

**Theorem 10.22 (66.4, Cauchy integral formula, classical version)**

Let $C$ be a simple closed piecewise-differentiable curve in $\mathbb{C}$, $B$ the bounded component of $\mathbb{R}^2 - C$, and $F$ analytic on an open set $\Omega \supset B \cup C$. For $a \in B$,

$$F(a) = \pm \frac{1}{2\pi i} \int_C \frac{F(z)}{z-a}\,dz,$$

with $+$ if $C$ is oriented counterclockwise and $-$ otherwise.

**Proof.** General version (Ahlfors): if $F$ is analytic on a region $\Omega$, $f$ a piecewise-differentiable loop in $\Omega$ with $n(f,b) = 0$ for all $b \notin \Omega$, and $a \in \Omega - f(I)$, then $n(f,a)F(a) = \frac{1}{2\pi i}\int_f \frac{F(z)}{z-a}dz$. For a parametrization $f$ of $C$, every $b \notin \Omega$ is in the unbounded component, so $n(f,b) = 0$, and $n(f,a) = \pm 1$ for $a \in B$ (Theorem 10.20). ■

# Chapter 11. The Seifert–van Kampen Theorem

## 11.1 Direct sums of abelian groups

**Motivation.** Direct sums, characterized by a universal property, model the free product to come.

Abelian groups are written additively. Subgroups $G_\alpha$ ($\alpha \in J$) of $G$ **generate** $G$ if every $x \in G$ is a finite sum $\sum x_\alpha$, $x_\alpha \in G_\alpha$ (almost all 0); $G$ is their **direct sum**, $G = \bigoplus G_\alpha$, if this expression is unique. $G$ is **free abelian** with **basis** $\{a_\alpha\}$ if each $a_\alpha$ generates an infinite cyclic $G_\alpha$ and $G = \bigoplus G_\alpha$.

**Lemma 11.1 (67.1)**

If $G = \bigoplus G_\alpha$, then $G$ has the **extension property**: for every abelian group $H$ and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each $h_\alpha$. Conversely, if the $G_\alpha$ generate $G$ and the extension property holds, then $G = \bigoplus G_\alpha$.

**Proof.** If $G = \bigoplus G_\alpha$, $h(\sum x_\alpha) = \sum h_\alpha(x_\alpha)$ is well defined, a homomorphism, and forced. Conversely let $\sum x_\alpha = \sum y_\alpha$; for fixed $\beta$ extend $h_\beta = \mathrm{id} : G_\beta \to G_\beta$, $h_\alpha = 0$ ($\alpha \neq \beta$), and apply the extension: $x_\beta = y_\beta$. ■

**Corollary 11.2 (67.2)**

If $G = G_1 \oplus G_2$, $G_1 = \bigoplus_{\alpha \in J} H_\alpha$ and $G_2 = \bigoplus_{\beta \in K} H_\beta$ with $J \cap K = \emptyset$, then $G = \bigoplus_{\gamma \in J \cup K} H_\gamma$.

**Proof.** The $H_\gamma$ generate $G$; homomorphisms $h_\gamma$ extend to $G_1$, $G_2$, then to $G$, uniquely as the $H_\gamma$ generate. Apply Lemma 11.1. ■

**Corollary 11.3 (67.3)**

If $G = G_1 \oplus G_2$, then $G/G_2 \cong G_1$.

**Proof.** Extend $\mathrm{id} : G_1 \to G_1$ and $0 : G_2 \to G_1$ to $h : G \to G_1$, onto with kernel $G_2$. ■

**Theorem 11.4 (67.4)**

For any family $\{G_\alpha\}$ of abelian groups there are an abelian group $G$ and monomorphisms $i_\alpha : G_\alpha \to G$ with $G = \bigoplus i_\alpha(G_\alpha)$ (the **external direct sum**).

**Proof.** Take the tuples in $\prod G_\alpha$ with finitely many nonzero entries, and $i_\beta(x)$ the tuple with $x$ in coordinate $\beta$, 0 elsewhere; each element is uniquely a finite sum of such. ■

**Lemma 11.5 (67.5)**

If $i_\alpha : G_\alpha \to G$ are monomorphisms and $G = \bigoplus i_\alpha(G_\alpha)$, then for any abelian $H$ and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

**Proof.** Apply Lemma 11.1 to $h_\alpha \circ i_\alpha^{-1} : i_\alpha(G_\alpha) \to H$. ■

**Theorem 11.6 (67.6, uniqueness of direct sums)**

If $G = \bigoplus i_\alpha(G_\alpha)$ and $G' = \bigoplus i'_\alpha(G_\alpha)$ with monomorphisms $i_\alpha, i'_\alpha$, there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

**Proof.** Lemma 11.5 gives $\phi$ with $\phi i_\alpha = i'_\alpha$ and $\psi$ with $\psi i'_\alpha = i_\alpha$. Then $\psi\phi i_\alpha = i_\alpha$, so $\psi\phi = \mathrm{id}$ by uniqueness ($H = G$, $h_\alpha = i_\alpha$); similarly $\phi\psi = \mathrm{id}$. ■

**Lemma 11.7 (67.7)**

Let $\{a_\alpha\}$ generate the abelian group $G$. Then $G$ is free abelian with basis $\{a_\alpha\}$ iff for every abelian $H$ and elements $y_\alpha \in H$ there is a homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$; such $h$ is then unique.

**Proof.** Let $G_\alpha = \langle a_\alpha \rangle$. If the condition holds, $H = \mathbb{Z}$, $y_\alpha = 1$ shows $na_\alpha = 0 \Rightarrow n = 0$; homomorphisms $G_\alpha \to H$ correspond to choices of $y_\alpha$, so the extension property holds and $G = \bigoplus G_\alpha$ (Lemma 11.1). Conversely, $na_\alpha \mapsto ny_\alpha$ defines $h_\alpha$ on the infinite cyclic $G_\alpha$; extend by Lemma 11.1. Uniqueness: the $a_\alpha$ generate. ■

**Theorem 11.8 (67.8)**

If $G$ is free abelian with basis $\{a_1, \dots, a_n\}$, then $n$ is determined by $G$.

**Proof.** $2G = \{2x\}$ is the set of $\sum n_i a_i$ with all $n_i$ even, the kernel of the surjection $\sum n_i a_i \mapsto (n_i \bmod 2) \in (\mathbb{Z}/2)^n$. So $|G/2G| = 2^n$, which depends only on $G$. ■

$n$ is the **rank** of $G$.

## 11.2 Free products of groups

**Motivation.** The free product combines groups with no relations; van Kampen's theorem produces it.

Let $\{G_\alpha\}$ be subgroups generating a group $G$ with $G_\alpha \cap G_\beta = \{1\}$ for $\alpha \neq \beta$. A **word** is a finite sequence $(g_1, \dots, g_n)$ in $\bigcup G_\alpha$, representing $g_1 \cdots g_n$; it is **reduced** if no $g_i = 1$ and no consecutive $g_i, g_{i+1}$ lie in one $G_\alpha$. Every element has a reduced word (merge adjacent letters of one group, delete 1's); 1 has the empty word. $G$ is the **free product** $\prod^* G_\alpha$ if each element has only one reduced word. It suffices that no nonempty reduced word represent 1: if reduced $(g_1, \dots, g_n)$, $(h_1, \dots, h_m)$ represent the same element, $g_1 \cdots g_n h_m^{-1} \cdots h_1^{-1} = 1$; if $g_n, h_m$ lie in different groups this word is reduced and nonempty, if in one group and different, merging gives one, and if $g_n = h_m$ cancel and induct.

**Lemma 11.9 (68.1)**

If $G = \prod^* G_\alpha$, then for every group $H$ and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each $h_\alpha$.

**Proof.** For $x$ with reduced word $(g_1, \dots, g_n)$, $g_i \in G_{\alpha_i}$, set

$$h(x) = h_{\alpha_1}(g_1) \cdots h_{\alpha_n}(g_n),$$

well defined by uniqueness. The formula gives the same value on any word (merging or deleting 1's does not change the product), and concatenation represents $xy$, so $h(xy) = h(x)h(y)$. Uniqueness: the $G_\alpha$ generate. ■

**Theorem 11.10 (68.2)**

For any family $\{G_\alpha\}$ of groups there are a group $G$ and monomorphisms $i_\alpha : G_\alpha \to G$ such that $G$ is the free product of the subgroups $i_\alpha(G_\alpha)$ (the **external free product**).

**Proof.** Take the $G_\alpha - \{1\}$ disjoint, and $W$ the set of reduced words in their union (with the empty word). For $g \in G_\alpha$ define $\pi_g : W \to W$: $\pi_1 = \mathrm{id}$; for $g \neq 1$, $w = (g_1, \dots, g_n)$, $\pi_g(w) = (g, g_1, \dots, g_n)$ if $w$ is empty or $g_1 \notin G_\alpha$; $(gg_1, g_2, \dots, g_n)$ if $g_1 \in G_\alpha$, $gg_1 \neq 1$; $(g_2, \dots, g_n)$ if $gg_1 = 1$. By cases $\pi_g \pi_{g'} = \pi_{gg'}$ for $g, g' \in G_\alpha$ (both change only a first letter in $G_\alpha$). So $\pi_g$ is a bijection with inverse $\pi_{g^{-1}}$, and $i_\alpha(g) = \pi_g$ is a homomorphism into the permutations of $W$, injective as $\pi_g(\emptyset) = (g)$. Let $G = \langle \pi_g \rangle$. For a nonempty reduced word, $\pi_{g_1} \cdots \pi_{g_n}(\emptyset) = (g_1, \dots, g_n) \neq \emptyset$, so it does not represent 1; likewise $i_\alpha(G_\alpha) \cap i_\beta(G_\beta) = \{1\}$. ∎

**Lemma 11.11 (68.3)**

If $i_\alpha : G_\alpha \to G$ are monomorphisms and $G$ is the free product of the $i_\alpha(G_\alpha)$, then for every group $H$ and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

**Proof.** Apply Lemma 11.9 to $h_\alpha \circ i_\alpha^{-1}$. ∎

**Theorem 11.12 (68.4, uniqueness of free products)**

Let $i_\alpha : G_\alpha \to G$ and $i'_\alpha : G_\alpha \to G'$ be monomorphisms whose images generate $G$ and $G'$, both with the extension property of Lemma 11.11. Then there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

**Proof.** As in Theorem 11.6: the extensions $\phi, \psi$ satisfy $\psi\phi i_\alpha = i_\alpha$, so $\psi\phi = \mathrm{id}$; likewise $\phi\psi = \mathrm{id}$. ∎

**Lemma 11.13 (68.5)**

If homomorphisms $i_\alpha : G_\alpha \to G$ satisfy the extension condition of Lemma 11.11, then each $i_\alpha$ is a monomorphism and $G$ is the free product of the $i_\alpha(G_\alpha)$.

**Proof.** Extending $h_\beta = \mathrm{id} : G_\beta \to G_\beta$, $h_\alpha$ trivial ($\alpha \neq \beta$), gives $h \circ i_\beta = \mathrm{id}$, so $i_\beta$ is injective. Let $G'$, $i'_\alpha$ be the external free product (Theorem 11.10). Extensions give $\phi : G \to G'$, $\phi i_\alpha = i'_\alpha$, and $\psi : G' \to G$, $\psi i'_\alpha = i_\alpha$; uniqueness for $G$ gives $\psi\phi = \mathrm{id}_G$, and $\phi\psi = \mathrm{id}$ as the $i'_\alpha(G_\alpha)$ generate $G'$. So $\phi$ is an isomorphism carrying $i_\alpha(G_\alpha)$ onto $i'_\alpha(G_\alpha)$. ∎

**Corollary 11.14 (68.6)**

If $G = G_1 * G_2$ with $G_1 = \prod^*_{\alpha \in J} H_\alpha$, $G_2 = \prod^*_{\beta \in K} H_\beta$ and $J \cap K = \emptyset$, then $G = \prod^*_{\gamma \in J \cup K} H_\gamma$.

**Proof.** Homomorphisms $h_\gamma$ extend to $G_1$, $G_2$, then to $G$, uniquely as the $H_\gamma$ generate. Apply Lemma 11.13 to the inclusions. ∎

**Theorem 11.15 (68.7)**

Let $G = G_1 * G_2$, $N_i \lhd G_i$, and $N$ the least normal subgroup of $G$ containing $N_1$ and $N_2$. Then $G/N \cong (G_1/N_1) * (G_2/N_2)$.

**Proof.** Let $F = (G_1/N_1) * (G_2/N_2)$. $G_i \to G_i/N_i \to F$ extend to an onto $\phi : G \to F$ killing $N_1, N_2$, hence $N$, inducing $\bar{\phi} : G/N \to F$. $gN_i \mapsto gN$ ($N_i \subset N$) extend to $\psi : F \to G/N$. $\psi\bar{\phi}$ and $\bar{\phi}\psi$ fix the generators $gN$, $gN_i$ ($g \in G_i$), so are the identity. ∎

**Corollary 11.16 (68.8)**

If $N$ is the least normal subgroup of $G_1 * G_2$ containing $G_1$, then $(G_1 * G_2)/N \cong G_2$.

**Proof.** Take $N_1 = G_1$, $N_2 = \{1\}$ in Theorem 11.15. ■

**Lemma 11.17 (68.9)**

The least normal subgroup $N$ of $G$ containing $S \subset G$ is generated by the conjugates of elements of $S$.

**Proof.** The subgroup $N'$ generated by these conjugates contains $S$ and is normal, as

$$g(x_1 \cdots x_k)g^{-1} = (gx_1g^{-1}) \cdots (gx_kg^{-1})$$

and a conjugate of a conjugate is a conjugate. Every normal subgroup containing $S$ contains $N'$, so $N = N'$. ■

## 11.3 Free groups

**Motivation.** Free groups are free products of copies of $\mathbb{Z}$; every group is a quotient of one.

$G$ is a **free group** with **system of free generators** $\{a_\alpha\}$ if each $a_\alpha$ generates an infinite cyclic subgroup $G_\alpha$ and $G = \prod^* G_\alpha$.

**Lemma 11.18 (69.1)**

If $G$ is free on $\{a_\alpha\}$, then for every group $H$ and elements $y_\alpha \in H$ there is a unique homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$. Conversely, if the $a_\alpha$ generate $G$ and this extension condition holds, $G$ is free on $\{a_\alpha\}$.

**Proof.** If $G$ is free, $a_\alpha^n \mapsto y_\alpha^n$ defines homomorphisms on the infinite cyclic $G_\alpha$; extend by Lemma 11.9. Conversely, $H = \mathbb{Z}$, $y_\alpha = 1$ shows $G_\alpha$ is infinite cyclic; homomorphisms $G_\alpha \to H$ correspond to choices of $y_\alpha$, so the condition of Lemma 11.11 holds for the inclusions, and Lemma 11.13 applies. ■

**Theorem 11.19 (69.2)**

If $G = G_1 * G_2$ with $G_1, G_2$ free on $\{a_\alpha\}_{\alpha \in J}$ and $\{a_\alpha\}_{\alpha \in K}$, $J \cap K = \emptyset$, then $G$ is free on $\{a_\alpha\}_{\alpha \in J \cup K}$.

**Proof.** Corollary 11.14. ■

If $G$ is free on $\{a_\alpha\}$, $\{r_\beta\} \subset G$, and $N$ is the least normal subgroup containing the $r_\beta$, a group isomorphic to $G/N$ has the **presentation** with generators $a_\alpha$ and relations $r_\beta$. The **commutator subgroup** $[G, G]$ is generated by all $[x, y] = xyx^{-1}y^{-1}$.

**Lemma 11.20 (69.3)**

$[G, G]$ is normal and $G/[G, G]$ is abelian. Every homomorphism $h : G \to H$ into an abelian group has $[G, G] \subset \ker h$, so it induces $k : G/[G, G] \to H$.

**Proof.** $g[x,y]g^{-1} = [gxg^{-1}, gyg^{-1}]$; $xy(yx)^{-1} = [x,y]$, so cosets commute; if $H$ is abelian, $h([x,y]) = 1$. ∎

**Theorem 11.21 (69.4)**

If $G$ is free on $\{a_\alpha\}$, then $G/[G,G]$ is free abelian with basis the cosets $[a_\alpha]$.

**Proof.** The cosets generate. For $y_\alpha$ in an abelian $H$, Lemma 11.18 gives $h(a_\alpha) = y_\alpha$, inducing $k([a_\alpha]) = y_\alpha$ (Lemma 11.20). Apply Lemma 11.7. ∎

**Corollary 11.22 (69.5)**

If $G$ is free on $n$ generators, every system of free generators of $G$ has $n$ elements.

**Proof.** $G/[G,G]$ is free abelian of rank the number of free generators (Theorem 11.21), unique by Theorem 11.8. (An infinite system would make $G/[G,G]$ not finitely generated.) ∎

## 11.4 The Seifert–van Kampen theorem

**Motivation.** $\pi_1(U \cup V)$ is $\pi_1(U) * \pi_1(V)$ with the images of $\pi_1(U \cap V)$ identified.

**Theorem 11.23 (70.1, Seifert–van Kampen theorem)**

Let $X = U \cup V$ with $U, V$ open and $U$, $V$, $U \cap V$ path connected, $x_0 \in U \cap V$. Let $i_1, i_2$ be induced by the inclusions of $U \cap V$ into $U$ and $V$, and $j_1, j_2$ by the inclusions of $U$ and $V$ into $X$. If $\phi_1 : \pi_1(U, x_0) \to H$ and $\phi_2 : \pi_1(V, x_0) \to H$ are homomorphisms with $\phi_1 \circ i_1 = \phi_2 \circ i_2$, there is a unique homomorphism $\Phi : \pi_1(X, x_0) \to H$ with $\Phi \circ j_1 = \phi_1$ and $\Phi \circ j_2 = \phi_2$.

**Proof.** Uniqueness: the images of $j_1, j_2$ generate $\pi_1(X, x_0)$ (Theorem 9.38). Write $[f]_U, [f]_V, [f]$ for path classes in $U$, $V$, $X$.

**Step 1.** For a loop $f$ at $x_0$ in $U$ or $V$ let $\rho(f) = \phi_1([f]_U)$ or $\phi_2([f]_V)$ (equal for $f$ in $U \cap V$, as $\phi_1 i_1 = \phi_2 i_2$). Then (1) $[f]_U = [g]_U$ or $[f]_V = [g]_V$ implies $\rho(f) = \rho(g)$;

(2) if $f, g$ both lie in $U$ or both in $V$, $\rho(f * g) = \rho(f)\rho(g)$.

**Step 2.** For $x \in X$ choose a path $\alpha_x$ from $x_0$ to $x$: constant if $x = x_0$, in $U \cap V$ if $x \in U \cap V$, in $U$ (resp. $V$) if $x$ lies only in $U$ (resp. $V$). For a path $f$ in $U$ or $V$ from $x$ to $y$, $L(f) = \alpha_x * f * \bar{\alpha}_y$ is a loop at $x_0$ in $U$ or $V$; let $\sigma(f) = \rho(L(f))$. $\sigma$ extends $\rho$ ($L(f) \simeq_p f$ for loops at $x_0$), satisfies (1), and (2) whenever $f * g$ is defined, as $L(f) * L(g) \simeq_p L(f * g)$ in $U$ or $V$.

**Step 3.** For a path $f$ in $X$ choose $0 = s_0 < \cdots < s_n = 1$ with each $[s_{i-1}, s_i]$ mapped into $U$ or $V$, $f_i$ the reparametrized pieces, and $\tau(f) = \sigma(f_1) \cdots \sigma(f_n)$. Adding a point replaces $\sigma(f_i)$ by $\sigma(f_i')\sigma(f_i'')$, equal since $f_i \simeq_p f_i' * f_i''$ in $U$ or $V$; so $\tau$ is well defined (common refinements) and extends $\sigma$.

**Step 4.** If $f \simeq_p g$ in $X$, then $\tau(f) = \tau(g)$. Special case: a path homotopy $F$ with a subdivision $s_0 < \cdots < s_n$ such that each $R_i = [s_{i-1}, s_i] \times I$ maps into $U$ or $V$. Let $\beta_i(t) = F(s_i, t)$ ($\beta_0, \beta_n$ constant). In $R_i$ the bottom-right and left-top edge paths are path homotopic (convexity), so $f_i * \beta_i \simeq_p \beta_{i-1} * g_i$ in $U$ or $V$ and $\sigma(f_i)\sigma(\beta_i) = \sigma(\beta_{i-1})\sigma(g_i)$. Also $\sigma(\beta_0) = \sigma(\beta_n) = 1$ ($\beta * \beta \simeq_p \beta$ for constant $\beta$). Multiplying, $\tau(f) = \sigma(f_1) \cdots \sigma(f_n) = \sigma(g_1) \cdots \sigma(g_n) = \tau(g)$. General case: subdivide $I \times I$ into rectangles $[s_{i-1}, s_i] \times [t_{j-1}, t_j]$ mapped into $U$ or $V$ and apply the special case to consecutive $f_j(s) = F(s, t_j)$.

**Step 5.** $\tau(f * g) = \tau(f)\tau(g)$: a subdivision for $f * g$ containing $\frac{1}{2}$ splits into subdivisions for $f$ and $g$.

**Step 6.** $\Phi([f]) = \tau(f)$ is a well-defined homomorphism (Steps 4, 5), and for a loop $f$ in $U$, $\Phi(j_1[f]_U) = \tau(f) = \rho(f) = \phi_1([f]_U)$; similarly for $V$. ■

**Theorem 11.24 (70.2, Seifert–van Kampen theorem, classical version)**

Under the same hypotheses, let $j : \pi_1(U, x_0) * \pi_1(V, x_0) \to \pi_1(X, x_0)$ extend $j_1$ and $j_2$. Then $j$ is surjective, and its kernel is the least normal subgroup $N$ containing all elements $i_1(g)^{-1} i_2(g)$, $g \in \pi_1(U \cap V, x_0)$.

**Proof.** Surjectivity is Theorem 9.38. As $j_1 i_1 = j_2 i_2$, $N \subset \ker j$, so $j$ induces an epimorphism $k : F/N \to \pi_1(X, x_0)$, $F$ the free product. Let $H = F/N$, $\phi_1, \phi_2$ the inclusions into $F$ followed by projection; $\phi_1 i_1(g) = i_1(g)N = i_2(g)N = \phi_2 i_2(g)$, so Theorem 11.23 gives $\Phi$ with $\Phi j_i = \phi_i$. For $g \in \pi_1(U, x_0)$, $\Phi k(gN) = \Phi j_1(g) = gN$, similarly for $V$; these cosets generate $H$, so $\Phi \circ k = \mathrm{id}$, $k$ is injective, and $\ker j = N$. ■

**Corollary 11.25 (70.3)**

If moreover $U \cap V$ is simply connected, then $\pi_1(U, x_0) * \pi_1(V, x_0) \cong \pi_1(X, x_0)$.

**Corollary 11.26 (70.4)**

If $V$ is simply connected, then $\pi_1(U, x_0)/N \cong \pi_1(X, x_0)$, where $N$ is the least normal subgroup of $\pi_1(U, x_0)$ containing the image of $i_1$.

**Proof.** In Theorem 11.24, $N$ is trivial in the first case; in the second, the free product is $\pi_1(U, x_0)$ and $i_1(g)^{-1} i_2(g) = i_1(g)^{-1}$. ■

**Example 11.1**

A theta space $X = A \cup B \cup C$ (arcs meeting in their end points) has free $\pi_1$ of rank 2: for interior points $a \in A$, $b \in B$, $U = X - a$ and $V = X - b$ deformation retract onto the circles $B \cup C$, $A \cup C$, and $U \cap V$ is contractible; apply Corollary 11.25.

## 11.5 The fundamental group of a wedge of circles

**Motivation.** A wedge of $n$ circles has $\pi_1$ free on $n$ generators.

A Hausdorff space $X$ that is the union of subspaces $S_1, \ldots, S_n$, each homeomorphic to $S^1$, with $S_i \cap S_j = \{p\}$ for $i \neq j$, is the **wedge** of these circles.

**Theorem 11.27 (71.1)**

If $X$ is the wedge of $S_1, \ldots, S_n$ and $f_i$ is a loop in $S_i$ representing a generator of $\pi_1(S_i, p)$, then $\pi_1(X, p)$ is free with free generators $[f_1], \ldots, [f_n]$.

**Proof.** Induction on $n$; $n = 1$ is clear. Choose $q_i \in S_i - p$, $W_i = S_i - q_i$, $U = S_1 \cup W_2 \cup \cdots \cup W_n$, $V = W_1 \cup S_2 \cup \cdots \cup S_n$; $U$, $V$, $U \cap V = W_1 \cup \cdots \cup W_n$ are open and path connected. The open arcs $W_i$ deformation retract onto $p$, and these fit together (pasting lemma on the closed $W_i \times I$) to a deformation retraction of $U \cap V$ onto $p$; so $\pi_1(X, p) \cong \pi_1(U, p) * \pi_1(V, p)$ (Corollary 11.25). Similarly $U$, $V$ deformation retract onto $S_1$, $S_2 \cup \cdots \cup S_n$, so $\pi_1(U, p) = \langle [f_1] \rangle \cong \mathbb{Z}$ and, by induction, $\pi_1(V, p)$ is free on $[f_2], \ldots, [f_n]$. Apply Theorem 11.19. ■

A union $X$ of subspaces $X_\alpha$ has the topology **coherent** with them if $C \subset X$ is closed whenever each $C \cap X_\alpha$ is closed in $X_\alpha$. If $X$ is the union of $S_\alpha \cong S^1$ ($\alpha \in J$) with $S_\alpha \cap S_\beta = \{p\}$ for $\alpha \neq \beta$, and coherent with them, $X$ is the **wedge** of the $S_\alpha$ (for finite $J$ this agrees with the earlier definition).

**Lemma 11.28 (71.2)**

A wedge $X$ of circles $S_\alpha$ is normal, and every compact subspace lies in the union of finitely many $S_\alpha$.

**Proof.** Points are closed by coherence. Let $A, B$ be disjoint closed, $p \notin B$. In each $S_\alpha$ choose disjoint open $U_\alpha \supset \{p\} \cup (A \cap S_\alpha)$, $V_\alpha \supset B \cap S_\alpha$. $U = \bigcup U_\alpha$, $V = \bigcup V_\alpha$ are disjoint, $U \cap S_\alpha = U_\alpha$ (all $U_\beta$ contain $p$, the only common point), $V \cap S_\alpha = V_\alpha$ ($p \notin V_\beta$), so $U, V$ are open. For compact $C$ choose $x_\alpha \in C \cap (S_\alpha - p)$ where possible; every subset of $D = \{x_\alpha\}$ meets each $S_\alpha$ in at most one point, so is closed; $D$ is closed discrete in the limit point compact $C$, hence finite. ■

**Theorem 11.29 (71.3)**

If $X$ is the wedge of circles $S_\alpha$ and $f_\alpha$ represents a generator of $\pi_1(S_\alpha, p)$, then $\pi_1(X, p)$ is free with free generators $\{[f_\alpha]\}$.

**Proof.** Let $G_\alpha$ be the image of $\pi_1(S_\alpha, p)$. Loops and path homotopies lie in finite unions of circles (Lemma 11.28). Hence, by Theorem 11.27 in finite wedges: the $G_\alpha$ generate; $\pi_1(S_\alpha, p) \to G_\alpha$ is injective, so $G_\alpha \cong \mathbb{Z}$; and a nonempty reduced word representing 1 would already be trivial in a finite wedge, a contradiction. ■

**Example 11.2**

The **infinite earring** $X \subset \mathbb{R}^2$, the union of the circles $C_n$ of radius $1/n$ centred at $(1/n, 0)$, is not a wedge: its intersection with the positive $x$-axis meets each $C_n$ in one point but is not closed. Indeed the $[f_n]$ do not generate $\pi_1(X, p)$: let $g$ run through $f_n$ on $[1/(n+1), 1/n]$, $g(0) = p$, a continuous loop. For $N > n$, the retraction of $X$ onto $C_N$ kills $[f_1], \ldots, [f_n]$ but sends $[g]$ to the generator $[f_N]$, so $[g] \notin \langle [f_1], \ldots, [f_n] \rangle$ for every $n$.

**Lemma 11.30 (71.4)**

For every index set $J$ there is a wedge of circles $S_\alpha$, $\alpha \in J$.

**Proof.** With $J$ discrete, let $X$ be the quotient of $E = S^1 \times J$ collapsing $P = b_0 \times J$ to $p$, $\pi$ the quotient map, $S_\alpha = \pi(S^1 \times \alpha)$. For $C$ closed in $S^1 \times \alpha$, $\pi^{-1}\pi(C)$ is $C$ or $C \cup P$, closed, so $\pi(C)$ is closed; hence $S_\alpha$ is closed and $\pi : S^1 \times \alpha \to S_\alpha$ is a homeomorphism. If each $D \cap S_\alpha$ is closed, so is each $\pi^{-1}(D) \cap (S^1 \times \alpha)$, hence $\pi^{-1}(D)$ and $D$: the topology is coherent. ■

## 11.6 Adjoining a two-cell

**Motivation.** Attaching a disc along a loop adds that loop as a relation.

**Theorem 11.31 (72.1)**

Let $X$ be Hausdorff, $A \subset X$ closed and path connected, and $h : B^2 \to X$ continuous, mapping Int $B^2$ bijectively onto $X - A$ and $S^1$ into $A$. Let $p \in S^1$, $a = h(p)$ and $k = h|S^1 : (S^1, p) \to$

$(A, a)$. Then $i_* : \pi_1(A, a) \to \pi_1(X, a)$ induced by inclusion is surjective, and its kernel is the least normal subgroup containing $k_*(\pi_1(S^1, p))$.

**Proof.**

**Step 1.** With $x_0 = h(0)$, $A$ is a deformation retract of $U = X - x_0$. $C = h(B^2)$ is closed; $\pi = h : B^2 \to C$ gives a closed map $\pi \times \mathrm{id} : B^2 \times I \to C \times I$ (compact to Hausdorff), a quotient map, as is its restriction to the saturated open $(B^2 - 0) \times I$. So the radial deformation retraction of $B^2 - 0$ onto $S^1$ induces one of $C - x_0$ onto $\pi(S^1)$; extend it by fixing $A$ (pasting lemma: the closed sets $C - x_0$ and $A$ of $U$ meet in the fixed $\pi(S^1)$). So $\pi_1(A, a) \cong \pi_1(U, a)$ (Theorem 9.33), and it suffices that $\pi_1(U, a) \to \pi_1(X, a)$ is onto with kernel the least normal subgroup containing $[g]$, $g = h \circ f$, $f$ a generator of $\pi_1(S^1, p)$.

**Step 2.** $V = X - A = h(\operatorname{Int} B^2)$ is open; $\pi | \operatorname{Int} B^2$ is a bijective quotient map (saturated open restriction), a homeomorphism, so $V$ is simply connected. $U \cap V = V - x_0 \cong \operatorname{Int} B^2 - 0$ is path connected with $\pi_1 \cong \mathbb{Z}$, and $U$ is path connected. For $b \in U \cap V$, Corollary 11.26: $\pi_1(U, b) \to \pi_1(X, b)$ is onto with kernel the least normal subgroup containing the image of $\pi_1(U \cap V, b)$.

**Step 3.** Let $q$ be the midpoint of $0p$, $b = h(q)$, $f_0$ a generating loop of $\operatorname{Int} B^2 - 0$ at $q$, $g_0 = h \circ f_0$, $\gamma$ the segment from $q$ to $p$, $\delta = h \circ \gamma$. The $\hat{\delta}$ commute with inclusion $U \to X$, so $\pi_1(U, a) \to \pi_1(X, a)$ is onto with kernel the least normal subgroup containing $\hat{\delta}([g_0])$. $\bar{\gamma} * f_0 * \gamma$ generates $\pi_1(B^2 - 0, p)$, so is path homotopic there to $f$ or $\bar{f}$; applying $h$, $\hat{\delta}([g_0]) = [g]^{\pm 1}$ in $\pi_1(U, a)$. ■

A space homeomorphic to $B^2$ is a **2-cell**; $X$ is obtained from $A$ by **adjoining a 2-cell**.

## 11.7 The fundamental groups of the torus and the dunce cap

**Motivation.** Polygon models give $\pi_1(T) = \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$ and $\pi_1$ of the dunce cap.

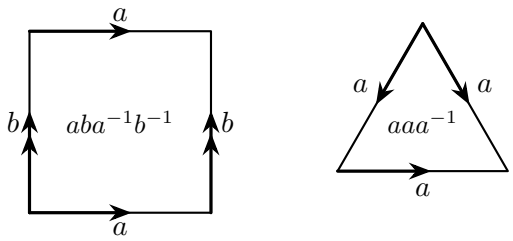



**Figure.** *Left: the torus, boundary word* $aba^{-1}b^{-1}$, $\pi_1 \cong \mathbb{Z} \times \mathbb{Z}$. *Right: a dunce cap, boundary word* $aaa^{-1}$, $\pi_1 = \langle a \mid a^2a^{-1} \rangle$ *trivial.*

**Theorem 11.32 (73.1)**

The fundamental group of the torus has a presentation with two generators $\alpha, \beta$ and one relation $\alpha\beta\alpha^{-1}\beta^{-1}$.

**Proof.** Let $h : I^2 \to T = S^1 \times S^1$ be the restriction of $p \times p$, $A = h(\operatorname{Bd} I^2)$, a wedge of two circles, $a = h(0, 0)$; Theorem 11.31 applies ($I^2 \cong B^2$). With $a_0(t) = (t, 0)$, $b_0(t) = (0, t)$, $a_1(t) = (t, 1)$, $b_1(t) = (1, t)$, $\alpha = h \circ a_0$, $\beta = h \circ b_0$ freely generate $\pi_1(A, a)$ (Theorem 11.27). $f = a_0 * (b_1 * (\bar{a}_1 * \bar{b}_0))$ generates $\pi_1(\operatorname{Bd} I^2, p)$, and $h \circ f = \alpha * (\beta * (\bar{\alpha} * \bar{\beta}))$. Apply Theorem 11.31. ■

**Corollary 11.33 (73.2)**

The fundamental group of the torus is free abelian of rank 2.

**Proof.** Let $G$ be free on $\alpha, \beta$, $N$ the least normal subgroup containing the commutator $\alpha\beta\alpha^{-1}\beta^{-1}$; so $N \subset [G,G]$. $G/N$ is generated by the commuting $\alpha N, \beta N$, so abelian and $[G,G] \subset N$. Hence $G/N = G/[G,G]$, free abelian of rank 2 (Theorem 11.21). ■

For $n > 1$ let $r$ be rotation of $S^1$ by $2\pi/n$. The **$n$-fold dunce cap** $X$ is the quotient of $B^2$ identifying each $x \in S^1$ with $r(x), \dots, r^{n-1}(x)$. The quotient map $\pi$ is closed: $\pi^{-1}\pi(C) = C \cup r(C_0) \cup \cdots \cup r^{n-1}(C_0)$, $C_0 = C \cap S^1$. So $X$ is compact, and Hausdorff by the next lemma.

**Lemma 11.34 (73.3)**

If $\pi : E \to X$ is a closed quotient map and $E$ is normal, so is $X$.

**Proof.** Points of $X$ are closed. For disjoint closed $A, B \subset X$ take disjoint open $U \supset \pi^{-1}(A)$, $V \supset \pi^{-1}(B)$. $U_0 = X - \pi(E - U)$ and $V_0 = X - \pi(E - V)$ are open ($\pi$ closed), contain $A$, $B$, and are disjoint: $x \in U_0 \Rightarrow \pi^{-1}(x) \subset U$, $x \in V_0 \Rightarrow \pi^{-1}(x) \subset V$. ■

The 2-fold dunce cap is $P^2$: $B^2 \to$ upper hemisphere $\to P^2$ is a closed continuous surjection, injective on $\operatorname{Int} B^2$, identifying $x$ with $-x$ on $S^1$.

**Theorem 11.35 (73.4)**

The fundamental group of the $n$-fold dunce cap is cyclic of order $n$.

**Proof.** Let $h : B^2 \to X$ be the quotient map, $A = h(S^1)$, $p = (1,0)$, $a = h(p)$. $h$ maps the arc of $S^1$ from $p$ to $r(p)$ onto $A$, identifying only its end points, so $A$ is a simple closed curve and, for $\gamma(t) = (\cos 2\pi t/n, \sin 2\pi t/n)$, $\alpha = h \circ \gamma$ generates $\pi_1(A,a) \cong \mathbb{Z}$. $f = \gamma * (r \circ \gamma) * \cdots * (r^{n-1} \circ \gamma)$ generates $\pi_1(S^1, p)$, and as $h \circ r^m = h$ on $S^1$, $h \circ f = \alpha^n$. By Theorem 11.31, $\pi_1(X,a) \cong \mathbb{Z}/n\mathbb{Z}$. ■

# Chapter 12. Classification of Surfaces

## 12.1 Fundamental groups of surfaces

**Motivation.** Gluing polygon edges in pairs gives every compact surface and a presentation of its $\pi_1$.

A **polygonal region** $P$ is the convex hull of points $p_0, \dots, p_{n-1}$ ($n \geq 3$) on a circle in counterclockwise order; its **edges** are the segments $p_{i-1}p_i$ ($p_n = p_0$). An **orientation** of an edge is a choice of initial point; a **labelling** assigns a label to each edge. Pasting edges with the same label by the positive linear maps respecting orientations gives a quotient space $X$. This is recorded by a **labelling scheme** $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$: the $k$-th edge has label $a_{i_k}$ and is oriented counterclockwise if $\epsilon_k = +1$, clockwise if $\epsilon_k = -1$; several disjoint regions are handled alike. For example, $aba^{-1}b^{-1}$ gives the torus, $abab$ the projective plane (the 2-fold dunce cap), and $aa^{-1}bb^{-1}$ the sphere.

**Theorem 12.1 (74.1)**

The space $X$ obtained from finitely many polygonal regions by pasting edges according to a labelling scheme is compact Hausdorff.

**Proof.** Compactness is clear. By Lemma 11.34 (the regions are normal) it suffices that $\pi$ is closed. For closed $C$ and an edge $e$, $\pi^{-1}\pi(C) \cap e$ is the union of $C \cap e$ and the $h_i(C \cap e_i)$, $h_i : e_i \to e$ the pasting homeomorphisms; this is compact, hence closed, and $\pi^{-1}\pi(C)$ is the union of $C$ and these finitely many sets. ∎

**Theorem 12.2 (74.2)**

Let $X$ be obtained from a polygonal region $P$ by a scheme $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$, with quotient map $\pi$. If $\pi$ maps all vertices to one point $x_0$ and $a_1, \dots, a_k$ are the distinct labels, then $\pi_1(X, x_0)$ is the quotient of the free group on $\alpha_1, \dots, \alpha_k$ by the least normal subgroup containing $(\alpha_{i_1})^{\epsilon_1} \cdots (\alpha_{i_n})^{\epsilon_n}$.

**Proof.** $A = \pi(\operatorname{Bd} P)$ is a wedge of $k$ circles, one per label. For each $i$ let $f_i$ be the counterclockwise linear path along an edge labelled $a_i$, $g_i = \pi \circ f_i$; the $[g_i]$ (or inverses) freely generate $\pi_1(A, x_0)$ (Theorem 11.27). $P \cong B^2$, $\pi$ maps $\operatorname{Int} P$ bijectively onto $X - A$, and for the loop $f$ once counterclockwise around $\operatorname{Bd} P$, $\pi \circ f = g_{i_1}^{\epsilon_1} * \cdots * g_{i_n}^{\epsilon_n}$ ($g^{-1} = \bar{g}$). Theorem 11.31 gives the result. ∎

The **$n$-fold torus** $T_n = T\# \cdots \#T$ is obtained from a $4n$-gon by the scheme

$$(a_1 b_1 a_1^{-1} b_1^{-1}) \cdots (a_n b_n a_n^{-1} b_n^{-1});$$

for $n = 2$ it is the double torus of Theorem 9.46. The **$m$-fold projective plane** $P_m$ is obtained from a $2m$-gon by $(a_1 a_1) \cdots (a_m a_m)$ ($m > 1$); set $P_1 = P^2$.

**Theorem 12.3 (74.3)**

$\pi_1(T_n, x_0)$ is the quotient of the free group on $\alpha_1, \beta_1, \dots, \alpha_n, \beta_n$ by the least normal subgroup containing $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, where $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$.

**Proof.** By Theorem 12.2 it suffices that all vertices are identified. In a block $aba^{-1}b^{-1}$ on $v_0, \dots, v_4$, $a$ pastes $v_0 \sim v_3$, $v_1 \sim v_2$ and $b$ pastes $v_1 \sim v_4$, $v_2 \sim v_3$; consecutive blocks share a vertex. ∎

**Theorem 12.4 (74.4)**

$\pi_1(P_m, x_0)$ is the quotient of the free group on $\alpha_1, \ldots, \alpha_m$ by the least normal subgroup containing $\alpha_1^2\alpha_2^2 \cdots \alpha_m^2$.

**Proof.** In a block $aa$ on $v_0, v_1, v_2$, $v_0 \sim v_1 \sim v_2$; consecutive blocks share a vertex. Apply Theorem 12.2. ■

## 12.2 Homology of surfaces

**Motivation.** Abelianizing $\pi_1$ gives an easily compared invariant that separates the standard surfaces.

For path-connected $X$ let $\pi_1 = \pi_1(X, x_0)$. For paths $\alpha, \beta$ from $x_0$ to $x_1$, $\hat{\alpha}$ and $\hat{\beta}$ induce the same isomorphism $\pi_1(X, x_0)/[\pi_1, \pi_1] \to \pi_1(X, x_1)/[\pi_1, \pi_1]$: with $g = \alpha * \bar{\beta}$, $\hat{g}([f]) = [g]^{-1}[f][g]$ has the coset of $[f]$. This group, determined up to canonical isomorphism, is the **first homology group** $H_1(X)$; homeomorphisms induce isomorphisms of $H_1$.

**Theorem 12.5 (75.1)**

Let $N \lhd F$ and $q : F \to F/N$, $p : F \to F/[F, F]$ the projections. Then $p$ induces an isomorphism $\phi : q(F)/[q(F), q(F)] \to p(F)/p(N)$.

**Proof.** Let $r : p(F) \to p(F)/p(N)$, $s : q(F) \to q(F)/[q(F), q(F)]$ be projections. $r \circ p$ kills $N$, inducing $u : q(F) \to p(F)/p(N)$, which kills commutators (abelian target), inducing $\phi$. Likewise $s \circ q$ induces $v : p(F) \to q(F)/[q(F), q(F)]$, and $v \circ p = s \circ q$ kills $N$, so $v$ induces $\psi$. As $\phi(sq(x)) = rp(x)$ and $\psi(rp(x)) = sq(x)$, they are inverse. ■

**Corollary 12.6 (75.2)**

Let $F$ be free on $\alpha_1, \ldots, \alpha_n$, $N$ the least normal subgroup containing $x$, and $G = F/N$. Then $G/[G, G]$ is the quotient of the free abelian group $F/[F, F]$ (basis $p(\alpha_i)$) by the subgroup generated by $p(x)$.

**Proof.** $N$ is generated by the conjugates of $x$ (Lemma 11.17), all with image $p(x)$, so $p(N) = \langle p(x) \rangle$. Apply Theorem 12.5 and Theorem 11.21. ■

**Theorem 12.7 (75.3)**

$H_1(T_n)$ is free abelian of rank $2n$.

**Proof.** By Theorem 12.3 and Corollary 12.6, $H_1(T_n)$ is free abelian on $\alpha_i, \beta_i$ modulo the image of $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, which is 0. ■

**Theorem 12.8 (75.4)**

The torsion subgroup $T(P_m)$ of $H_1(P_m)$ has order 2, and $H_1(P_m)/T(P_m)$ is free abelian of rank $m - 1$.

**Proof.** $H_1(P_m)$ is free abelian on $\alpha_1, \ldots, \alpha_m$ modulo $\langle 2\gamma \rangle$, $\gamma = \alpha_1 + \cdots + \alpha_m$. As $\alpha_1, \ldots, \alpha_{m-1}, \gamma$ is another basis, $H_1(P_m) \cong \mathbb{Z}^{m-1} \times \mathbb{Z}/2$. ■

**Theorem 12.9 (75.5)**

The surfaces $S^2$, $T_1, T_2, \ldots$, $P_1, P_2, \ldots$ are topologically distinct.

**Proof.** Their $H_1$ are 0, $\mathbb{Z}^{2n}$, $\mathbb{Z}^{m-1} \times \mathbb{Z}/2$, pairwise non-isomorphic (by rank, Theorem 11.8, and torsion). ■

## 12.3 Cutting and pasting

**Motivation.** Cutting and regluing reduce any scheme to a normal form.

Let $P$ have vertices $p_0, \ldots, p_{n-1}$ and $1 < k < n-1$. Cutting along $p_0p_k$ gives $Q_1$ (vertices $p_0, \ldots, p_k$) and $Q_2$ (vertices $p_k, \ldots, p_n = p_0$); translating $Q_1$ off $Q_2$, $P$ is homeomorphic to the quotient of $Q_1 \sqcup Q_2$ pasting the two copies of $p_0p_k$ linearly. Conversely, two regions pasted along one edge each form one region (move the vertices of one onto the circle of the other).

**Theorem 12.10 (76.1)**

Let $X$ be obtained from $m$ regions by the scheme $y_0y_1, w_2, \ldots, w_m$, and let $c$ be a label not appearing in it. If $y_0$ and $y_1$ have length at least 2, then $X$ is also obtained from $m+1$ regions by the scheme $y_0c^{-1}, cy_1, w_2, \ldots, w_m$. Conversely, the second scheme gives the same space as the first.

**Proof.** Cut the first region along the diagonal $p_0p_k$, $k$ the length of $y_0$, labelling both copies $c$ (counterclockwise in the second piece, clockwise in the first). A composite of quotient maps is a quotient map, so pasting $c$ first and then the other edges gives the same space. ■

The following **elementary operations** do not change the space; schemes related by them are **equivalent**, and each has an inverse.

(i) *Cut*: replace $y_0y_1$ by $y_0c^{-1}$ and $cy_1$ ($c$ new, $y_0, y_1$ of length $\geq 2$).

(ii) *Paste*: replace $y_0c^{-1}$ and $cy_1$ by $y_0y_1$ ($c$ appearing nowhere else).

(iii) *Relabel*: replace all occurrences of a label by a new label, or change the sign of all exponents of one label.

(iv) *Permute*: replace $y_0y_1$ by $y_1y_0$ (renumber the vertices).

(v) *Flip*: replace $(a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$ by $(a_{i_n})^{-\epsilon_n} \cdots (a_{i_1})^{-\epsilon_1}$ (reflect the region).

(vi) *Cancel*: replace $y_0aa^{-1}y_1$ by $y_0y_1$ if $a$ appears nowhere else and $y_0, y_1$ have length $\geq 2$. Cut to $y_0ab$ and $b^{-1}a^{-1}y_1$, merge $a, b$ into one edge $c$ in each (the composite pasting is linear), giving $y_0c$, $c^{-1}y_1$; paste.

(vii) *Uncancel*: the reverse of (vi).

**Example 12.1**

The Klein bottle $aba^{-1}b$ is $P_2$: cut to $abc^{-1}$, $ca^{-1}b$; permute the first to $bc^{-1}a$; flip the second to $b^{-1}ac^{-1}$, permute to $ac^{-1}b^{-1}$; paste along $b$: $ac^{-1}c^{-1}a$; permute and relabel to $aabb$.

## 12.4 The classification theorem

**Motivation.** Compact connected surfaces are $S^2$, $T_n$ or $P_m$, all distinct.

A scheme is **proper** if each label appears exactly twice (preserved by elementary operations). A proper one-region scheme is of **torus type** if each label appears once with each exponent, of **projective type** otherwise. Brackets only group letters; $y^{-1}$ is the formal inverse of a word $y$.

**Lemma 12.11 (77.1)**

If $w = [y_0]a[y_1]a[y_2]$ is proper (some $y_i$ possibly empty), then $w \sim aa[y_0y_1^{-1}y_2]$.

**Proof.**

**Step 1.** $y_0$ empty, $w = ay_1ay_2$. If $y_1$ is empty, done. If $y_2$ is empty, flip to $a^{-1}y_1^{-1}a^{-1}$, permute and relabel to $aay_1^{-1}$. Otherwise cut into $ay_1c^{-1}$, $cay_2$; flip the first to $cy_1^{-1}a^{-1}$, permute the second to $ay_2c$; paste along $a$: $cy_1^{-1}y_2c$; permute and relabel to $aay_1^{-1}y_2$.

**Step 2.** $y_0$ nonempty. If $y_1, y_2$ are empty, permute. Otherwise cut into $y_0ac^{-1}$, $cy_1ay_2$; flip and permute the first to $y_0^{-1}ca^{-1}$, permute the second to $ay_2cy_1$; paste along $a$: $y_0^{-1}cy_2cy_1$. Permute to $cy_2cy_1y_0^{-1}$; by Step 1, $\sim ccy_2^{-1}y_1y_0^{-1}$; flip to $y_0y_1^{-1}y_2c^{-1}c^{-1}$; permute and relabel to $aay_0y_1^{-1}y_2$. ■

**Corollary 12.12 (77.2)**

A scheme of projective type is equivalent to one of the same length of the form $(a_1a_1)(a_2a_2)\cdots(a_ka_k)w_1$ with $k \geq 1$ and $w_1$ empty or of torus type.

**Proof.** Some label occurs twice with the same exponent, say $+1$; Lemma 12.11 gives $w \sim aaw_1$. If $w_1$ is not of torus type, write $w = aaz_0bz_1bz_2$ and apply the lemma with $y_0 = aaz_0$: $w \sim bbaaz_0z_1^{-1}z_2$. Each step lengthens the initial block of pairs, so the process ends. ■

One more consequence of the operations:

**Step E.** *If $a$ does not occur in $B, D_1, D_2$, then the cyclic scheme $aBa^{-1}D_1D_2$ is equivalent to $aBa^{-1}D_2D_1$.* Take $D_1, D_2$ nonempty. Permute to $Ba^{-1}D_1D_2a$, cut into $Ba^{-1}D_1c^{-1}$, $cD_2a$, permute to $D_1c^{-1}Ba^{-1}$, $acD_2$, paste along $a$: $D_1c^{-1}BcD_2$; permute and relabel ($c \to a^{-1}$): $aBa^{-1}D_2D_1$. Writing $aBa^{-1}D$ cyclically as $a^{-1}DaB$ and relabelling, the same holds for $B$: *either block between $a$ and $a^{-1}$ may be cyclically rotated.*

**Lemma 12.13 (77.3)**

Let $w = w_0w_1$ be proper with $w_1$ of torus type containing no two adjacent terms with the same label. Then $w \sim w_0w_2$ with $w_2$ of the same length as $w_1$ and of the form $aba^{-1}b^{-1}w_3$, $w_3$ empty or of torus type.

**Proof.** Let $a$ be a label of $w_1$ whose two occurrences are closest; they are not adjacent; say $a$ comes first. Let $b$ occur between them, with exponent $+1$ after relabelling; $b^{-1}$ is not between $a$ and $a^{-1}$. If it follows $a^{-1}$, $w_1 = y_1aPbQa^{-1}Rb^{-1}y_5$; if it precedes $a$, exchange $a$ and $b$ (changing exponents). Cyclically $w = aPbQa^{-1}Rb^{-1}S$, $S = y_5w_0y_1$. By Step E: rotate $PbQ$ (between $a$, $a^{-1}$) to $bQP$: $ab(QP)a^{-1}Rb^{-1}S$; rotate $(QP)a^{-1}R$ (between $b$, $b^{-1}$) to $a^{-1}RQP$: $aba^{-1}(RQP)b^{-1}S$; rotate $(RQP)b^{-1}S$ (after $a^{-1}$) to $b^{-1}SRQP$: $aba^{-1}b^{-1}T$, $T = SRQP$. For any $T = T_1T_2$, $aba^{-1}b^{-1}T_1T_2 \sim aba^{-1}b^{-1}T_2T_1$: rotate $T_1T_2a$ (between $b^{-1}$ and $b$) to $T_2aT_1$ and repeat the three rotations. With $T_1 = y_5w_0$, $T_2 = y_1RQP$, permuting gives $w_0aba^{-1}b^{-1}w_3$, $w_3 = y_1RQPy_5$ the remaining letters of $w_1$ with unchanged exponents, so empty or of torus type. ■

**Lemma 12.14 (77.4)**

If $w = w_0(cc)(aba^{-1}b^{-1})w_1$ is proper, then $w \sim w_0aabbccw_1$.

**Proof.** Permute to $cc[ab][ba]^{-1}[w_1w_0]$. By Lemma 12.11 backwards ($y_0 = ab$, $y_1 = ba$, $y_2 = w_1w_0$), this is $\sim [ab]c[ba]c[w_1w_0] = [a]b[c]b[acw_1w_0]$, by the lemma $\sim bb[ac^{-1}acw_1w_0] = [bb]a[c^{-1}]a[cw_1w_0]$, and again $\sim aa[bbccw_1w_0]$. Permute. ■

**Theorem 12.15 (77.5, classification theorem)**

A space obtained from a polygonal region by pasting its edges together in pairs is homeomorphic to $S^2$, to $T_n$, or to $P_m$.

**Proof.** $w$ is proper of length $\geq 4$. We show $w$ is equivalent to one of (1) $aa^{-1}bb^{-1}$ ($S^2$), (2) $abab$ ($P^2$), (3) $(a_1a_1)\cdots(a_ma_m)$, $m \geq 2$ ($P_m$), (4) $(a_1b_1a_1^{-1}b_1^{-1})\cdots(a_nb_na_n^{-1}b_n^{-1})$ ($T_n$). If $w$ has length $> 4$ and contains $aa^{-1}$, cancelling (after permuting so both remaining parts have length $\geq 2$) gives a shorter scheme of the same type.

**Step 1 (torus type).** Length 4: up to permuting and relabelling, $aa^{-1}bb^{-1}$ or $aba^{-1}b^{-1}$. Longer $w$: induction, assuming no cancellation is possible. Lemma 12.13 gives $w \sim aba^{-1}b^{-1}w_3$, $w_3$ nonempty of torus type without adjacent pair; apply the lemma with $w_0 = aba^{-1}b^{-1}$, and so on, reaching type (4).

**Step 2 (projective type).** Length 4: Corollary 12.12 gives $aabb$ (type 3) or $aay$, $y = b^{\pm1}b^{\mp1}$; writing $y = y_1^{-1}y_2$ with single letters, Lemma 12.11 gives $ay_1ay_2$, relabelled $abab$. Longer $w$: Corollary 12.12 gives $(a_1a_1)\cdots(a_ka_k)w_1$. If $w_1$ is empty, type (3); if it has an adjacent pair, cancel and induct. Otherwise Lemma 12.13 gives $(a_1a_1)\cdots(a_ka_k)aba^{-1}b^{-1}w_2$, and Lemma 12.14 turns this into $(a_1a_1)\cdots(a_{k-1}a_{k-1})aabba_ka_kw_2$, with a shorter torus part. Repeat. ■

## 12.5 Constructing compact surfaces

**Motivation.** Triangulations show every compact surface comes from a polygon.

Let $X$ be compact Hausdorff. A **curved triangle** in $X$ is a subspace $A$ with a homeomorphism $h : T \to A$ from a closed triangular region; edges and vertices of $A$ are images of those of $T$. A **triangulation** of $X$ is a finite collection of curved triangles covering $X$, two distinct ones meeting in nothing, a common vertex, or a common edge, in the last case with $h_j^{-1} \circ h_i$ linear on the edge of $T_i$. Every compact surface is triangulable (Radó); we assume this.

**Theorem 12.16 (78.1)**

A compact triangulable surface is homeomorphic to the space obtained from finitely many disjoint triangular regions by pasting their edges together in pairs.

**Proof.** Let $A_1, \ldots, A_n$ be a triangulation, $h_i : T_i \to A_i$, $T_i$ disjoint. $h : E = \bigsqcup T_i \to X$ is a closed continuous surjection, hence a quotient map. It suffices that (a) each edge of each $A_i$ is an edge of exactly one other triangle, and (b) if $A_i \cap A_j$ is a vertex $v$, a chain of triangles with vertex $v$, consecutive ones sharing an edge, joins $A_i$ to $A_j$; then $h$ identifies exactly the points identified by the edge pastings.

**Step 1 (b).** Call triangles with vertex $v$ equivalent if so joined. If there were two classes, let $B$ be the union of one, $C$ of the rest; $B \cap C = \{v\}$. For a neighbourhood $W$ of $v$ missing the triangles not containing $v$, $W - v$ is the union of the disjoint relatively closed nonempty $(W \cap B) - v$, $(W \cap C) - v$, so disconnected. But $v$ has a disc neighbourhood, hence arbitrarily small $W$ with $W - v$ connected.

**Step 2 (at least one other triangle).** Let $x$ be interior to an edge of a triangular region $T$. Its $\epsilon$-neighbourhoods $W$ in $T$ have $W - x$ star-shaped about an interior point, hence simply connected. But if $x$ had a neighbourhood $U \cong B$ (open unit disc, $x \mapsto 0$), then for every neighbourhood $V \subset U$

of $x$ and $B_\epsilon \subset V$, the inclusion $B_\epsilon - 0 \to B - 0$ is homotopic to the homeomorphism $y \mapsto y/\epsilon$, so $\pi_1(V-0) \to \pi_1(B-0) \cong \mathbb{Z}$ is onto and $V - 0$ is not simply connected. Now if an edge $e$ of $A_i$ lay in no other triangle, a point interior to $e$ would have a neighbourhood in $X$ inside $A_i$, and a disc neighbourhood in $X$ would give one in $T_i$.

**Step 3 (at most one).** Let $Y \subset \mathbb{R}^3$ be the union of $k \geq 3$ triangles meeting pairwise in a common edge $e$, $x = 0$ interior to $e$. The union $A$ of the other edges is $k$ arcs with common end points; it retracts onto three of them (map each extra arc onto one of the three, fixing end points), a theta space with nonabelian $\pi_1$ (Example 11.1), so $\pi_1(A)$ is nonabelian. $A$ is a deformation retract of $Y-0$ (radial projection from 0 in each triangle). For a neighbourhood $W$ of 0, choose $\epsilon$ with $Y' = \epsilon Y \subset W$; $Y' - 0 \to Y - 0$ is homotopic to the homeomorphism $y \mapsto y/\epsilon$, so $\pi_1(W-0) \to \pi_1(Y-0)$ is onto and $\pi_1(W-0)$ is nonabelian (base point in $Y'$). A disc neighbourhood $W$ would have $\pi_1(W-0) \cong \mathbb{Z}$. If $e$ were an edge of three or more triangles, a point interior to $e$ would have a neighbourhood in $X$ inside their union, homeomorphic to such a $Y$. ■

**Theorem 12.17 (78.2)**

A compact connected triangulable surface is homeomorphic to a space obtained from one polygonal region by pasting its edges together in pairs.

**Proof.** By Theorem 12.16, $X$ comes from triangular regions by a proper scheme. While two different regions have edges with the same label, flip one if needed and paste (operation (ii)). If more than one region remained, each would give a nonempty clopen subset of $X$, contradicting connectedness. ■

By Theorems 12.17, 12.15 and 12.9, every compact connected surface is homeomorphic to exactly one of $S^2$, $T_n$, $P_m$.

## 12.6 Selected exercises

**Motivation.** The presentations and $H_1$ on concrete surfaces, and the list rewritten with $P^2$ and the Klein bottle.

**Exercise 12.1 (§74, Ex. 6–7; nonabelian surface groups)**

(a) Show that $\pi_1(T_n)$ is not abelian for $n > 1$.
(b) Show that $\pi_1(P_m)$ is not abelian for $m > 1$.

**Solution.**

**Step 1.** (Mapping a presented group.) Let $G = F/N$, $F$ free on $\gamma_i$, $N$ the least normal subgroup containing $r$. For $y_i$ in a group $H$ let $\phi : F \to H$, $\phi(\gamma_i) = y_i$ (Lemma 11.18). If $\phi(r) = e$, then $N \subset \ker \phi$ and $\phi$ induces $G \to H$. Images of abelian groups are abelian.

**Step 2.** In $S_3$, $(1,2)(1,3)$ and $(1,3)(1,2)$ are the two different 3-cycles, so $\langle (1,2), (1,3) \rangle$ is nonabelian.

(a) By Theorem 12.3, $r = [\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$. Send $\alpha_1, \beta_1 \mapsto (1,2)$, $\alpha_2, \beta_2 \mapsto (1,3)$, others to $e$; each $[x, x]$ is trivial, so $r \mapsto e$, and by Steps 1–2 $\pi_1(T_n)$ has a nonabelian image.

(b) By Theorem 12.4, $r = \alpha_1^2 \cdots \alpha_m^2$. Send $\alpha_1 \mapsto (1,2)$, $\alpha_2 \mapsto (1,3)$, others to $e$; transpositions have order 2, so $r \mapsto e$; conclude as in (a). ■

**Exercise 12.2 (§76, Ex. 1; the surface $P^2\#T$)**

Let $X = P^2\#T$ be the space obtained from a hexagon by the scheme $aabcb^{-1}c^{-1}$. Calculate $H_1(X)$ and decide which surface of Theorem 12.9 it can be.

**Solution.**

**Step 1.** (All vertices go to one point.) Let the $k$th edge run from $p_{k-1}$ to $p_k$ ($p_6 = p_0$). The $a$-edges give $p_0 \sim p_1 \sim p_2$; the $b$-edges ($p_2 \to p_3$, reversed $p_5 \to p_4$) give $p_2 \sim p_5$, $p_3 \sim p_4$; the $c$-edges ($p_3 \to p_4$, reversed $p_0 \to p_5$) give $p_3 \sim p_0$, $p_4 \sim p_5$.

**Step 2.** By Theorem 12.2 the relator is $\alpha^2\beta\gamma\beta^{-1}\gamma^{-1}$; by Corollary 12.6, $H_1(X)$ is free abelian on $\alpha, \beta, \gamma$ modulo $\langle 2\alpha + \beta + \gamma - \beta - \gamma\rangle = \langle 2\alpha\rangle$:

$$H_1(X) \cong \mathbb{Z}/2 \times \mathbb{Z} \times \mathbb{Z},$$

torsion of order 2, rank 2.

**Step 3.** Homeomorphic spaces have isomorphic $H_1$. $H_1(S^2) = 0$ and $H_1(T_n) \cong \mathbb{Z}^{2n}$ are torsion free (Theorem 12.7); $H_1(P_m)$ has torsion of order 2 and rank $m-1$ (Theorem 12.8). So $X$ can only be $P_3$; Exercise 12.3 shows it is. ■

**Exercise 12.3 (§77, Ex. 1(a); another form of the list)**

Write $K$ for the Klein bottle, and $T_n\#P^2$, $T_n\#K$ for the spaces given by the schemes $w_n cc$ and $w_n ccdd$, where $w_n = (a_1b_1a_1^{-1}b_1^{-1})\cdots(a_nb_na_n^{-1}b_n^{-1})$. Show that a space obtained by pasting the edges of a polygonal region in pairs is homeomorphic to exactly one of $S^2$, $P^2$, $K$, $T_n$, $T_n\#P^2$, $T_n\#K$ ($n \geq 1$).

**Solution.**

**Step 1.** (A squared letter absorbs a torus block.) For $k \geq 1$ and a word $u$, applying Lemma 12.14 $k$ times, each time to the pair $(cc)(a_ib_ia_i^{-1}b_i^{-1})$ after the letters already moved,

$$u\,(cc)\,w_k \sim u\,(a_1a_1)(b_1b_1)\cdots(a_ka_k)(b_kb_k)\,(cc).$$

**Step 2.** ($T_n\#P^2 \cong P_{2n+1}$.) Permute $w_n cc$ to $(cc)w_n$; Step 1 with $u$ empty gives $2n+1$ squares of distinct letters, the scheme of $P_{2n+1}$.

**Step 3.** ($T_n\#K \cong P_{2n+2}$.) Permute $w_n ccdd$ to $(cc)(dd)w_n$; Step 1 with $u = cc$ and $dd$ for $cc$ gives $2n+2$ squares: $P_{2n+2}$.

**Step 4.** (Existence.) By Theorem 12.15 the space is $S^2$, some $T_n$, or some $P_m$; $P_1 = P^2$, $P_2 \cong K$ (Example 12.1), and for $m \geq 3$, $P_m \cong T_n\#P^2$ ($n = (m-1)/2$, $m$ odd) or $T_n\#K$ ($n = (m-2)/2$, $m$ even) by Steps 2–3.

**Step 5.** (Uniqueness.) The list is $S^2$, $P_1$, $P_2$, $T_n$, $P_{2n+1}$, $P_{2n+2}$ ($n \geq 1$), pairwise distinct standard surfaces, hence non-homeomorphic (Theorem 12.9).

In particular $P^2\#T \cong P_3$, as in Exercise 12.2. ■

# Chapter 13. Classification of Covering Spaces

Throughout, a covering map $p : E \to B$ has $E$, $B$ path connected and locally path connected. For $p(e_0) = b_0$ let $H_0 = p_*(\pi_1(E, e_0)) \cong \pi_1(E, e_0)$ (Theorem 9.17).

## 13.1 Equivalence of covering spaces

**Motivation.** Covering spaces are classified by subgroups of $\pi_1$, as in Galois theory.

Covering maps $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p = p' \circ h$ (an **equivalence**).

**Lemma 13.1 (79.1, general lifting lemma)**

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f : Y \to B$ continuous with $f(y_0) = b_0$, where $Y$ is path connected and locally path connected. Then $f$ has a lifting $\tilde{f} : Y \to E$ with $\tilde{f}(y_0) = e_0$ iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0))$. Such a lifting is unique.

**Proof.** If $\tilde{f}$ exists, $f_* = p_*\tilde{f}_*$.

Uniqueness: for a path $\alpha$ from $y_0$ to $y_1$, $\tilde{f} \circ \alpha$ is the lifting of $f \circ \alpha$ from $e_0$, so $\tilde{f}(y_1)$ is its end point.

Existence: let $\tilde{f}(y_1)$ be the end point of the lifting of $f \circ \alpha$ from $e_0$. Well defined: for another path $\beta$, lift $f \circ \alpha$ to $\gamma$ from $e_0$ and $f \circ \bar{\beta}$ to $\delta$ from $\gamma(1)$; $\gamma * \delta$ lifts $f \circ (\alpha * \bar{\beta})$, whose class is in $f_*\pi_1(Y, y_0) \subset H_0$, so it is a loop (Theorem 9.17(c)); thus $\bar{\delta}$ lifts $f \circ \beta$ from $e_0$ and ends at $\gamma(1)$. Continuity at $y_1$: given $N \ni \tilde{f}(y_1)$, choose a path-connected evenly covered $U \ni f(y_1)$ whose slice $V_0$ through $\tilde{f}(y_1)$ lies in $N$, and a path-connected $W \ni y_1$ with $f(W) \subset U$. For a path $\delta$ in $W$ from $y_1$ to $y$, the lifting of $f \circ (\alpha * \delta)$ is $\gamma * ((p|V_0)^{-1} \circ f \circ \delta)$, ending in $V_0$. So $\tilde{f}(W) \subset V_0 \subset N$. ∎

**Theorem 13.2 (79.2)**

Let $p : E \to B$, $p' : E' \to B$ be covering maps with $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. Such $h$ is unique.

**Proof.** If $h$ exists, $h_*$ is an isomorphism and $p = p'h$ gives $H_0 = H_0'$. Conversely, Lemma 13.1 gives liftings $h : E \to E'$ of $p$, $h(e_0) = e_0'$, and $k : E' \to E$ of $p'$, $k(e_0') = e_0$. $k \circ h$ and $\mathrm{id}_E$ both lift $p$ and fix $e_0$, so are equal; likewise $h \circ k = \mathrm{id}$. Uniqueness of $h$ is that of liftings. ∎

**Lemma 13.3 (79.3)**

Let $e_0, e_1 \in p^{-1}(b_0)$ and $H_i = p_*\pi_1(E, e_i)$.
(a) If $\gamma$ is a path in $E$ from $e_0$ to $e_1$ and $\alpha = p \circ \gamma$, then $[\alpha]H_1[\alpha]^{-1} = H_0$.
(b) Conversely, if $H$ is conjugate to $H_0$, then $H = H_1$ for some $e_1 \in p^{-1}(b_0)$.

**Proof.** (a) If $[h] \in H_1$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at $e_1$; $\gamma * \tilde{h} * \bar{\gamma}$ is a loop at $e_0$ over $[\alpha][h][\alpha]^{-1}$, so $[\alpha]H_1[\alpha]^{-1} \subset H_0$. Using $\bar{\gamma}$, $[\alpha]^{-1}H_0[\alpha] \subset H_1$.

(b) If $H_0 = [\alpha]H[\alpha]^{-1}$, lift $\alpha$ to $\gamma$ from $e_0$, $e_1 = \gamma(1)$; by (a), $H_0 = [\alpha]H_1[\alpha]^{-1}$, so $H = H_1$. ∎

**Theorem 13.4 (79.4)**

$p$ and $p'$ are equivalent iff $H_0$ and $H_0'$ are conjugate in $\pi_1(B, b_0)$.

**Proof.** If $h$ is an equivalence and $e_1' = h(e_0)$, then $H_0 = p_*'\pi_1(E', e_1')$ (Theorem 13.2), conjugate to $H_0'$ (Lemma 13.3). Conversely, Lemma 13.3(b) gives $e_1'$ with $p_*'\pi_1(E', e_1') = H_0$, and Theorem 13.2 an equivalence $e_0 \mapsto e_1'$. ■

**Example 13.1**

$\pi_1(S^1) \cong \mathbb{Z}$ is abelian with subgroups $n\mathbb{Z}$, so every covering of $S^1$ is equivalent to $\mathbb{R} \to S^1$ (trivial subgroup) or $z \mapsto z^n$ (subgroup $n\mathbb{Z}$).

## 13.2 The universal covering space

**Motivation.** The simply connected covering lies above all others, like the splitting field.

If $E$ is simply connected, $p$ is a **universal covering**; it is unique up to equivalence (Theorem 13.4).

**Lemma 13.5 (80.1)**

Let $B$ be path connected and locally path connected, and $p : E \to B$ a covering map with $E$ not necessarily path connected. If $E_0$ is a path component of $E$, then $p|E_0 : E_0 \to B$ is a covering map.

**Proof.** $E$ is locally path connected, so $E_0$ and $p(E_0)$ are open. If $x \in \overline{p(E_0)}$, take a path-connected evenly covered $U \ni x$; some slice $V$ meets $E_0$, hence lies in $E_0$, so $x \in U = p(V) \subset p(E_0)$. Thus $p(E_0) = B$, and $(p|E_0)^{-1}(U)$ is the union of the slices meeting $E_0$, each in $E_0$ and mapped homeomorphically onto $U$. ■

**Lemma 13.6 (80.2)**

Let $p = r \circ q$ with $p : X \to Z$, $q : X \to Y$, $r : Y \to Z$ continuous (all spaces path connected and locally path connected).
(a) If $p$ and $r$ are covering maps, so is $q$.
(b) If $p$ and $q$ are covering maps, so is $r$.

**Proof.** Let $x_0 \in X$, $y_0 = q(x_0)$.

(a) $q$ is onto: for a path $\tilde{\alpha}$ in $Y$ from $y_0$ to $y$, lift $r \circ \tilde{\alpha}$ through $p$ to $\tilde{\tilde{\alpha}}$ from $x_0$; $q \circ \tilde{\tilde{\alpha}}$ lifts $r \circ \tilde{\alpha}$ through $r$ from $y_0$, so equals $\tilde{\alpha}$ and ends at $y$. Given $y$, take a path-connected $U \ni r(y)$ evenly covered by $p$ and $r$, and the slice $V \ni y$ of $r^{-1}(U)$. Each slice $U_\alpha$ of $p^{-1}(U)$ is connected, so $q$ maps it into one slice of $r^{-1}(U)$; $q^{-1}(V)$ is the union of those $U_\alpha$ mapped into $V$, and on each $q = (r|V)^{-1} \circ (p|U_\alpha)$ is a homeomorphism onto $V$.

(b) Let $U \ni z$ be path connected and evenly covered by $p$, and $V_\beta$ the path components of $r^{-1}(U)$, open and disjoint. The slices $U_\alpha$ of $p^{-1}(U)$ are its path components, and $q$ maps each into one $V_\beta$; so $q^{-1}(V_\beta)$ is a union of slices containing some $U_\alpha$ ($q$ onto). By Theorem 9.10, $q^{-1}(V_\beta) \to V_\beta$ is a covering map with path component $U_\alpha$, so $q|U_\alpha : U_\alpha \to V_\beta$ is onto (Lemma 13.5); it is open and injective (as $p|U_\alpha$ is), hence a homeomorphism, and $r|V_\beta = (p|U_\alpha) \circ (q|U_\alpha)^{-1}$ is a homeomorphism onto $U$. ■

**Theorem 13.7 (80.3)**

If $p : E \to B$ is a covering map with $E$ simply connected, then for any covering map $r : Y \to B$ there is a covering map $q : E \to Y$ with $r \circ q = p$.

**Proof.** Lemma 13.1 applies trivially and gives $q$ with $r \circ q = p$; Lemma 13.6(a) shows $q$ is a covering map. ■

**Lemma 13.8 (80.4)**

If $p : E \to B$ is a covering map with $E$ simply connected and $p(e_0) = b_0$, then $b_0$ has a neighbourhood $U$ such that $i_* : \pi_1(U, b_0) \to \pi_1(B, b_0)$ is trivial.

**Proof.** Let $U$ be evenly covered, $\tilde{U}$ the slice containing $e_0$. A loop $f$ in $U$ lifts to a loop in $\tilde{U}$, nulhomotopic in $E$; applying $p$ gives a path homotopy from $f$ to a constant. ■

**Example 13.2**

The infinite earring $X$ (Example 11.2) is path connected and locally path connected but has no universal covering: every neighbourhood $U$ of the origin contains some $C_n$, a retract of $X$, so $\pi_1(C_n) \to \pi_1(X)$ is injective and $i_* : \pi_1(U) \to \pi_1(X)$ is nontrivial.

## 13.3 Covering transformations

**Motivation.** Deck transformations, the analogue of the Galois group, give $\pi_1(B)$ for the universal cover.

An equivalence of $p$ with itself is a **covering transformation**; they form the group $\mathcal{C}(E, p, B)$. The **normalizer** of $H$ in $G$ is $N(H) = \{g : gHg^{-1} = H\}$. Let $F = p^{-1}(b_0)$, $\Phi : \pi_1(B, b_0)/H_0 \to F$ the bijection from the lifting correspondence (Theorem 9.17), and $\Psi : \mathcal{C}(E, p, B) \to F$, $\Psi(h) = h(e_0)$, injective by uniqueness of liftings.

**Lemma 13.9 (81.1)**

The image of $\Psi$ equals $\Phi(N(H_0)/H_0)$.

**Proof.** Let $\gamma$ lift a loop $\alpha$ from $e_0$, $e_1 = \gamma(1) = \Phi([\alpha]H_0)$, $H_1 = p_*\pi_1(E, e_1)$. Some $h$ has $h(e_0) = e_1$ iff $H_1 = H_0$ (Theorem 13.2); as $[\alpha]H_1[\alpha]^{-1} = H_0$ (Lemma 13.3), iff $[\alpha] \in N(H_0)$. ■

**Theorem 13.10 (81.2)**

$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$ is a group isomorphism.

**Proof.** It is a bijection by Lemma 13.9. Let $h(e_0) = e_1$, $k(e_0) = e_2$, $\gamma, \delta$ paths from $e_0$ to $e_1, e_2$, $\alpha = p \circ \gamma$, $\beta = p \circ \delta$, so $\Phi([\alpha]H_0) = e_1$, $\Phi([\beta]H_0) = e_2$. $h \circ \delta$ lifts $\beta$ from $e_1$ to $e_3 = h(k(e_0))$, so $\gamma * (h \circ \delta)$ lifts $\alpha * \beta$ from $e_0$ to $e_3$: $\Phi([\alpha * \beta]H_0) = \Psi(h \circ k)$. ■

**Corollary 13.11 (81.3)**

$H_0$ is normal in $\pi_1(B, b_0)$ iff for all $e_1, e_2 \in p^{-1}(b_0)$ some covering transformation carries $e_1$ to $e_2$. Then $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)/H_0$.

**Proof.** If $H_0$ is normal, $\Psi$ is onto, giving $h_i(e_0) = e_i$; take $h_2 h_1^{-1}$. Conversely, transitivity makes $\Psi$ onto, so $N(H_0) = \pi_1(B, b_0)$. ■

**Corollary 13.12 (81.4)**

If $E$ is simply connected, $\mathcal{C}(E,p,B) \cong \pi_1(B,b_0)$.

If $H_0$ is normal, $p$ is **regular**, and $\mathcal{C}$ is transitive on every fibre $p^{-1}(b)$: for $x_1, x_2 \in p^{-1}(b)$ and a path $\beta$ from $b$ to $b_0$, lift $\beta$ from $x_1, x_2$ to paths ending at $e_1, e_2$, and let $h(e_1) = e_2$; $h$ carries the first lifting to a lifting of $\beta$ ending at $e_2$, by uniqueness (lift $\bar{\beta}$ from $e_2$) the second, so $h(x_1) = x_2$.

**Example 13.3**

Every covering of $S^1$ is regular; for $\mathbb{R} \to S^1$ the covering transformations are $x \mapsto x + n$. For the covering of the figure eight in Lemma 9.45, $\mathcal{C}$ is trivial: a loop around $A$ lifts to a non-loop from the origin but to a loop from every other fibre point on the $y$-axis, similarly for $B$, so every covering transformation fixes the origin.

For a group $G$ of homeomorphisms of $X$, the **orbit space** $X/G$ is the quotient by $x \sim g(x)$. The action is **properly discontinuous** if each $x$ has a neighbourhood $U$ with $g(U) \cap U = \emptyset$ for $g \neq e$; then $g_0(U) \cap g_1(U) = \emptyset$ for $g_0 \neq g_1$.

**Theorem 13.13 (81.5)**

Let $X$ be path connected and locally path connected and $G$ a group of homeomorphisms of $X$. The quotient map $\pi : X \to X/G$ is a covering map iff the action is properly discontinuous. In that case $\pi$ is regular and $G$ is its group of covering transformations.

**Proof.** $\pi$ is open: $\pi^{-1}\pi(U) = \bigcup_g g(U)$. If the action is properly discontinuous, $\pi^{-1}\pi(U)$ is the disjoint union of the open $g(U)$, each meeting each orbit at most once; $\pi|g(U)$ is a continuous open bijection onto $\pi(U)$, a homeomorphism. Conversely, if $\pi$ is a covering, let $U$ be the slice through $x$ over an evenly covered neighbourhood of $\pi(x)$; if $g \neq e$ and $u, g(u) \in U$, then $g(u) = u$ ($\pi|U$ injective), so $g$ is a covering transformation fixing a point, hence the identity (shown next), a contradiction. Each $g$ is a covering transformation; if $h$ is one with $h(x_1) = x_2$, then $g(x_1) = x_2$ for some $g$, and $h = g$ (Theorem 13.2). $G$ is transitive on fibres, so $\pi$ is regular (Corollary 13.11). ■

**Theorem 13.14 (81.6)**

If $p : X \to B$ is regular with group of covering transformations $G$, there is a homeomorphism $k : X/G \to B$ with $p = k \circ \pi$.

**Proof.** $p$ is constant on orbits, so induces a continuous $k$. $p$ is a quotient map (continuous, open, onto) and $\pi$ is constant on its fibres ($G$ transitive on fibres), so $\pi$ induces a continuous inverse $B \to X/G$. ■

**Example 13.4**

On the cylinder $X = S^1 \times I$, $h(x,t) = (-x,t)$ and $k(x,t) = (-x, 1-t)$ generate properly discontinuous $\mathbb{Z}/2$-actions; $X/\langle h \rangle$ is a cylinder, $X/\langle k \rangle$ the Möbius band. $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translations gives the torus; $\{\pm 1\}$ on $S^n$ gives $P^n$.

## 13.4 Existence of covering spaces

**Motivation.** If $B$ is semilocally simply connected, every subgroup of $\pi_1(B)$ comes from a covering.

$B$ is **semilocally simply connected** if each $b$ has a neighbourhood $U$ with $\pi_1(U,b) \to \pi_1(B,b)$ trivial; this is necessary for a universal covering (Lemma 13.8).

**Theorem 13.15 (82.1)**

Let $B$ be path connected, locally path connected and semilocally simply connected, $b_0 \in B$, and $H$ a subgroup of $\pi_1(B,b_0)$. There is a covering map $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ with $p_*\pi_1(E,e_0) = H$.

**Proof.**

**Step 1.** Let $\mathcal{P}$ be the set of paths in $B$ from $b_0$; $\alpha \sim \beta$ if $\alpha(1) = \beta(1)$ and $[\alpha * \bar{\beta}] \in H$ (an equivalence relation). Write $\alpha^\#$ for the class, $E$ for the set of classes, $p(\alpha^\#) = \alpha(1)$. Then:

(1) $[\alpha] = [\beta]$ implies $\alpha^\# = \beta^\#$;

(2) $\alpha^\# = \beta^\#$ implies $(\alpha * \delta)^\# = (\beta * \delta)^\#$ for any path $\delta$ from $\alpha(1)$, as $[(\alpha * \delta) * \overline{(\beta * \delta)}] = [\alpha * \bar{\beta}]$.

**Step 2.** For $\alpha \in \mathcal{P}$ and a path-connected neighbourhood $U$ of $\alpha(1)$ let $B(U,\alpha) = \{(\alpha * \delta)^\# : \delta$ a path in $U$ from $\alpha(1)\} \ni \alpha^\#$. If $\beta^\# \in B(U,\alpha)$, then $B(U,\beta) = B(U,\alpha)$: with $\beta^\# = (\alpha * \delta)^\#$, $(\beta * \gamma)^\# = (\alpha * (\delta * \gamma))^\# \in B(U,\alpha)$ by (2), and $\alpha^\# = (\beta * \bar{\delta})^\#$ gives the reverse. Hence if $\beta^\# \in B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$ and $V \subset U_1 \cap U_2$ is a path-connected neighbourhood of $\beta(1)$, $B(V,\beta) \subset B(U_1,\beta) \cap B(U_2,\beta) = B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$. So these sets form a basis.

**Step 3.** $p$ is continuous ($B(U,\alpha) \subset p^{-1}(W)$ when $U \subset W$) and open ($p(B(U,\alpha)) = U$).

**Step 4.** Let $U$ be a path-connected neighbourhood of $b_1$ with $\pi_1(U,b_1) \to \pi_1(B,b_1)$ trivial. $p^{-1}(U)$ is the union of the $B(U,\alpha)$, $\alpha(1) = b_1$: if $p(\beta^\#) \in U$, take $\delta$ in $U$ from $b_1$ to $\beta(1)$ and $\alpha = \beta * \bar{\delta}$; then $\beta^\# = (\alpha * \delta)^\#$. Distinct such sets are disjoint (Step 2). $p$ is injective on $B(U,\alpha)$: if $(\alpha * \delta_1)(1) = (\alpha * \delta_2)(1)$, $\delta_1 * \bar{\delta}_2$ is a loop in $U$ at $b_1$, nulhomotopic in $B$, so $[\alpha * \delta_1] = [\alpha * \delta_2]$. As a continuous open bijection, $p|B(U,\alpha)$ is a homeomorphism onto $U$.

**Step 5.** Let $e_0$ be the class of the constant path. For $\alpha \in \mathcal{P}$ and $c \in I$ let $\alpha_c(t) = \alpha(tc)$, $\tilde{\alpha}(c) = \alpha_c^\#$; it lifts $\alpha$ from $e_0$ to $\alpha^\#$. Continuity at $c$: a basic neighbourhood of $\tilde{\alpha}(c)$ is $B(U,\alpha_c)$, $U$ a path-connected neighbourhood of $\alpha(c)$; for $d$ near $c$, $\alpha([c,d]) \subset U$ and $[\alpha_d] = [\alpha_c * \delta]$, $\delta$ the reparametrized restriction of $\alpha$ to $[c,d]$ (or $[d,c]$, reversed), so $\tilde{\alpha}(d) \in B(U,\alpha_c)$.

**Step 6.** $E$ is path connected (Step 5) and locally path connected (locally homeomorphic to $B$), so $p$ is a covering map.

**Step 7.** For a loop $\alpha$ at $b_0$, its lifting from $e_0$ ends at $\alpha^\#$, which is $e_0$ iff $[\alpha] \in H$. By Theorem 9.17(c), $p_*\pi_1(E,e_0) = H$. ∎

**Corollary 13.16 (82.2)**

$B$ has a universal covering space iff it is path connected, locally path connected and semilocally simply connected.

**Proof.** Necessity: Lemma 13.8 (each point of $B$ is the image of a point of $E$).

Sufficiency: Theorem 13.15 with $H$ trivial. ∎

With Theorem 13.4: for such $B$, equivalence classes of coverings $\leftrightarrow$ conjugacy classes of subgroups of $\pi_1(B,b_0)$.

## 13.5 Selected exercises

**Motivation.** Applications of the lifting lemma to maps into $S^1$, and a criterion for proper discontinuity (Theorem 13.13).

**Exercise 13.1 (§79, Ex. 1; maps of spheres into the circle)**

Show that if $n > 1$, every continuous map $f : S^n \to S^1$ is nulhomotopic.

**Solution.**

**Step 1.** $S^n$ is path connected, locally path connected (locally homeomorphic to $\mathbb{R}^n$ by stereographic projection), and simply connected for $n \geq 2$ (Theorem 9.40).

**Step 2.** Let $p : \mathbb{R} \to S^1$ be the covering of Theorem 9.9, $x_0 \in S^n$, $e_0 \in p^{-1}(f(x_0))$. $f_*(\pi_1(S^n, x_0))$ is trivial, so lies in $p_*(\pi_1(\mathbb{R}, e_0))$, and Lemma 13.1 gives $\tilde{f} : S^n \to \mathbb{R}$ with $p \circ \tilde{f} = f$.

**Step 3.** $H(x,t) = p((1-t)\tilde{f}(x))$ is a homotopy from $f$ to the constant $p(0)$. ■

**Exercise 13.2 (§79, Ex. 2; maps into the circle)**

(a) Show that every continuous map $f : P^2 \to S^1$ is nulhomotopic.
(b) Show that the projection $q : T \to S^1$ of the torus $T = S^1 \times S^1$ onto its first factor is not nulhomotopic.

**Solution.**

(a) $P^2$ is a surface (Theorem 9.43), so path connected and locally path connected. $\pi_1(P^2, x_0)$ has order 2 (Corollary 9.44), so $f_*$ of it is a finite subgroup of $\pi_1(S^1) \cong \mathbb{Z}$ (Theorem 9.16), hence $\{0\}$. Steps 2–3 of Exercise 13.1 apply with $P^2$ for $S^n$.

(b)

**Step 1.** Let $j : S^1 \to T$, $j(z) = (z, b_0)$; then $q \circ j = \mathrm{id}_{S^1}$.

**Step 2.** A homotopy $H$ from $q$ to a constant gives the homotopy $H \circ (j \times \mathrm{id}_I)$ from $\mathrm{id}_{S^1}$ to a constant.

**Step 3.** By Lemma 9.20, $(\mathrm{id}_{S^1})_*$ would be trivial, but it is the identity of $\pi_1(S^1, b_0) \cong \mathbb{Z} \neq 0$. So $q$ is not nulhomotopic. ■

**Exercise 13.3 (§81, Ex. 4; free actions of finite groups)**

Let $X$ be Hausdorff and $G$ a finite group of homeomorphisms of $X$ acting fixed-point freely (no $g \neq e$ has a fixed point). Show that the action of $G$ is properly discontinuous.

**Solution.** Let $G = \{g_0 = e, g_1, \ldots, g_k\}$ and $x \in X$.

**Step 1.** The $g_i(x)$ are distinct: $g_i(x) = g_j(x)$ gives $g_j^{-1} g_i(x) = x$, so $g_j^{-1} g_i = e$, $i = j$.

**Step 2.** For $i \neq j$ choose disjoint open $W_{ij} \ni g_i(x)$, $W_{ji} \ni g_j(x)$, and let $V_i = \bigcap_{j \neq i} W_{ij}$: open, pairwise disjoint, $g_i(x) \in V_i$.

**Step 3.** $U = \bigcap_{i=0}^{k} g_i^{-1}(V_i)$ is an open neighbourhood of $x$ with $U \subset V_0$ and $g_i(U) \subset V_i$, so $g_i(U) \cap U = \emptyset$ for $g_i \neq e$. ■

# Chapter 14. Applications to Group Theory

Free groups are fundamental groups of graphs, so covering theory proves algebraic facts about them. (Not in the source PDF; results are named, not numbered.)

## 14.1 Covering spaces of a graph

**Motivation.** Covering spaces of graphs are again graphs.

A **linear graph** is a space $X$ that is the union of arcs $A_\alpha$ (**edges**), any two meeting in at most a common end point (a **vertex**), whose topology is coherent with the edges: $C \subset X$ is closed iff $C \cap A_\alpha$ is closed in $A_\alpha$ for every $\alpha$. A **subgraph** is a union of edges and vertices; it is closed. Fix for each edge a homeomorphism with $[0,1]$. For a subgraph $S$, the **open star** $\mathrm{St}(S)$ consists of $S$ and the points of edges not in $S$ at parameter distance less than $\frac{1}{2}$ from an end point lying in $S$; it is open by coherence. Since $I$ is compact Hausdorff, $X \times I$ is coherent with the $A_\alpha \times I$ (quotient map $\times$ identity of a locally compact Hausdorff space is a quotient map).

**Lemma 14.1 (graphs are nice)**

A linear graph $X$ is Hausdorff and locally path connected; every compact subset meets the interiors of only finitely many edges; and each subgraph $S$ is a deformation retract of $\mathrm{St}(S)$. In particular $X$ is semilocally simply connected.

**Proof.** Edge interiors and the stars $\mathrm{St}_\epsilon(v)$ (points within parameter distance $\epsilon$ of $v$) are open by coherence and path connected; they separate points and give local path connectedness. For compact $C$, pick one point of $C$ in each edge interior where possible; every subset of the resulting $D$ meets each edge in at most one point, so $D$ is closed discrete in $C$, hence finite. Pushing each half-open segment of $\mathrm{St}(S) - S$ linearly to its end in $S$ is continuous on each $(A_\alpha \cap \mathrm{St}(S)) \times I$, hence (coherence) a deformation retraction of $\mathrm{St}(S)$ onto $S$. Finally $\mathrm{St}(v)$ and small open arcs are contractible, so loops in them are nulhomotopic. ■

**Theorem 14.2 (coverings of graphs)**

If $p : E \to X$ is a covering map and $X$ a linear graph, then $E$ is a linear graph whose edges are the liftings of the edges of $X$, and $p$ maps each edge of $E$ homeomorphically onto an edge of $X$.

**Proof.** Parametrize an edge $A$ by an injective path $f : I \to A$. For $e \in p^{-1}(f(0))$ the lifting $\tilde{f}$ from $e$ is injective (as $p \circ \tilde{f} = f$ is), so its image is an arc mapped homeomorphically onto $A$. By uniqueness of liftings each point of $p^{-1}(A)$ lies on exactly one lifted arc; lifts of different edges meet at most in common end points over vertices, and two lifts of the same edge are disjoint. Coherence: let $D \subset E$ meet each lifted edge in a closed set, $e \in E$. Take an evenly covered $U \ni p(e)$ that is a small vertex star or open arc, with slice $V \ni e$. Each segment of $U$ along one edge is connected, so lifts into $V$ inside one lifted edge; hence $p(D \cap V)$ meets each segment in a closed set, so is closed in $U$ (coherent with the segments), and $D \cap V$ is closed in $V$. Thus $E - D$ is open. ■

## 14.2 The fundamental group of a graph

**Motivation.** Collapsing a maximal tree shows $\pi_1$ of a graph is free, one generator per edge outside the tree.

A **tree** is a connected linear graph containing no simple closed curve.

**Lemma 14.3 (finite trees)**

A finite tree with $n \geq 1$ edges has a vertex lying on only one edge; it deformation retracts onto a vertex; and if it has $V$ vertices then it has $V - 1$ edges.

**Proof.** Take a maximal arc $e_1 \cup \cdots \cup e_k$ of distinct edges (the tree is finite). Its last end point $w$ lies on no other edge, which would extend the arc or close a simple closed curve. Removing $e_k$ (but its other end) leaves a tree $T'$ with $n - 1$ edges and one fewer vertex, and collapsing $e_k$ linearly deformation retracts $T$ onto $T'$. Induct. ■

**Theorem 14.4 (trees are simply connected)**

Every tree $T$ is simply connected.

**Proof.** A loop $f$ in $T$ meets finitely many edge interiors (Lemma 14.1); the component $Y_0 \supset f(I)$ of the union of these edges is a finite tree, so $f$ is nulhomotopic in $Y_0 \subset T$ (Lemma 14.3). ■

**Lemma 14.5 (maximal trees)**

Every connected linear graph $X$ contains a maximal tree, and a maximal tree contains every vertex of $X$.

**Proof.** A union of a chain of trees is connected and has no simple closed curve (a compact curve lies in finitely many edges, hence in one member). Zorn gives a maximal tree $T$. If a vertex is not in $T$: were every edge with an end in $T$ to have both ends in $T$, the subgraph $W$ of $T$'s vertices and these edges would be clopen, so $W = X$, a contradiction. So some edge $A$ has exactly one end in $T$, and $T \cup A$ is a tree (a simple closed curve through the new end needs two edges there), contradicting maximality. ■

**Theorem 14.6 (fundamental group of a graph)**

Let $X$ be a connected linear graph, $T$ a maximal tree, $x_0$ a vertex, and $\{A_\alpha\}$ the edges not in $T$. For each $\alpha$ let $f_\alpha$ be a loop at $x_0$ that runs in $T$ to one end of $A_\alpha$, across $A_\alpha$, and back in $T$. Then $\pi_1(X, x_0)$ is free with free generators $\{[f_\alpha]\}$.

**Proof.**

**Step 1.** $X = T \cup A_1 \cup \cdots \cup A_n$, induction on $n$ ($n = 0$: Theorem 14.4). Let $Y = T \cup A_1 \cup \cdots \cup A_{n-1}$, $a \in \operatorname{int} A_n$, $p, q$ the ends of $A_n$, $\gamma$ the arc in $T$ from $p$ to $q$, $C = \gamma \cup A_n$ (a simple closed curve), $U = X - a$, $V = \operatorname{St}(C)$. $U$ deformation retracts onto $Y$ (push the halves of $A_n - a$ to $p, q$; pasting lemma on $Y \times I$, $(A_n - a) \times I$); $V$ onto $C$ and $U \cap V$ onto the arc $C - a$ (Lemma 14.1). Corollary 11.25 gives $\pi_1(X, p) \cong \pi_1(Y, p) * \pi_1(C, p)$; the first factor is free on the loops through $A_1, \ldots, A_{n-1}$ (induction), the second is $\mathbb{Z}$ on the loop around $C$, so Theorem 11.19 gives the claim at $p$; moving the base point along $T$ turns these loops into the $f_\alpha$.

**Step 2.** general case. A loop or path homotopy has compact image, so lies in $X_F = T \cup \bigcup_{\alpha \in F} A_\alpha$ for a finite $F$. Hence the $[f_\alpha]$ generate, and a nonempty reduced word in them equal to 1 (or a finite-order $[f_\alpha]$) would already occur in some $X_F$, contradicting Step 1. ■

**Corollary 14.7 (Euler characteristic)**

If $X$ is a finite connected linear graph with $V$ vertices and $E$ edges, $\pi_1(X)$ is free of rank $E - V + 1 = 1 - \chi(X)$, where $\chi(X) = V - E$.

**Proof.** A maximal tree contains all $V$ vertices (Lemma 14.5), hence has $V - 1$ edges (Lemma 14.3), leaving $E - V + 1$ edges outside it. ■

Consequently $\chi$ of a finite connected graph depends only on $\pi_1$ (by Corollary 11.22), and is unchanged by subdivision.

## 14.3 Subgroups of free groups

**Motivation.** A subgroup of $\pi_1$ of a graph is $\pi_1$ of a covering graph, so subgroups of free groups are free.

Subdividing each circle into three arcs makes a wedge of circles a linear graph; by Lemma 11.30 and Theorem 11.29 every free group is $\pi_1$ of one.

**Theorem 14.8 (Nielsen–Schreier)**

Every subgroup $H$ of a free group $F$ is free.

**Proof.** Let $X$ be a wedge of circles with $\pi_1(X, x_0) \cong F$. $X$ is path connected, locally path connected and semilocally simply connected (Lemma 14.1), so Theorem 13.15 gives a covering $p : E \to X$ with $p_*\pi_1(E, e_0) = H$. As $p_*$ is injective, $H \cong \pi_1(E, e_0)$. $E$ is a connected linear graph (Theorem 14.2), so $\pi_1(E, e_0)$ is free (Theorem 14.6). ■

**Theorem 14.9 (Schreier index formula)**

If $F$ is free of rank $n$ and $H$ has finite index $k$ in $F$, then $H$ is free of rank $k(n - 1) + 1$.

**Proof.** In the proof above, the fibre $p^{-1}(x_0)$ is in bijection with the right cosets of $H$ (Theorem 9.17), so $p$ is $k$-sheeted: each vertex and each edge of $X$ has exactly $k$ preimages. The subdivided wedge $X$ has $2n + 1$ vertices and $3n$ edges, so $\chi(X) = 1 - n$ and $\chi(E) = k(1 - n)$. By Corollary 14.7, $\pi_1(E)$ has rank $1 - \chi(E) = k(n - 1) + 1$. ■

**Example 14.1**

(a) $F$ free on $a, b$ is $\pi_1$ of the figure eight $X$. The integer grid in $\mathbb{R}^2$ covers $X$ (Example 9.3); the integer translations act transitively on fibres, so the covering is regular with $F/H \cong \mathbb{Z}^2$, $H = p_*\pi_1(E)$. As $F/H$ is abelian, $H \supset [F, F]$, and the surjection $\mathbb{Z}^2 \cong F/[F, F] \to F/H \cong \mathbb{Z}^2$ is injective (determinant $\pm 1$), so $H = [F, F]$. The comb ($x$-axis and all vertical lines) is a maximal tree with infinitely many edges outside, so $[F, F]$ is free of infinite rank.
(b) $\ker(F \to \mathbb{Z}/2)$, $a, b \mapsto 1$, has index 2, hence rank $2(2 - 1) + 1 = 3$: a subgroup of a free group can have larger rank.

## 14.4 Selected exercises

**Motivation.** Standard consequences of the theorems above (the source has no exercises here), with our own solutions.

**Exercise 14.1 (recognizing finite trees)**

Let $X$ be a finite connected linear graph with $V$ vertices and $E$ edges.
(a) Show that $E \geq V - 1$, with equality iff $X$ is a tree.
(b) Show that $X$ is a tree iff $\pi_1(X)$ is trivial.

**Solution [Supplement].**

(a) A maximal tree $T$ contains all $V$ vertices (Lemma 14.5), so has $V-1$ edges (Lemma 14.3; $V = 1$ if none). Hence $E \geq V - 1$; $X$ a tree $\Rightarrow T = X$, $E = V - 1$; $E = V - 1 \Rightarrow$ every edge lies in $T$, so $X = T$.

(b) Trees are simply connected (Theorem 14.4). If $\pi_1(X) = 1$, its rank $E-V+1$ is 0 (Corollary 14.7), so $X$ is a tree by (a). ■

**Exercise 14.2 (free groups are torsion free)**

Show that every element $g \neq e$ of a free group $F$ has infinite order.

**Solution [Supplement].**

**Step 1.** (Free on $S \neq \emptyset \Rightarrow$ infinite.) For $s \in S$, Lemma 11.18 gives $\phi : H \to \mathbb{Z}$ with $\phi(s) = 1$, $\phi(s') = 0$ $(s' \neq s)$; $\phi(s^k) = k$, so the $s^k$ are distinct.

**Step 2.** If $g \neq e$ had finite order, $\langle g \rangle$ would be finite, nontrivial, and free on some $S \neq \emptyset$ (Theorem 14.8), contradicting Step 1. ■

**Exercise 14.3 (free subgroups of every finite rank)**

Let $F$ be free on $\{a, b\}$. Show that for every $n \geq 1$, $F$ has a subgroup that is free of rank $n$.

**Solution [Supplement].**

**Step 1.** $(n = 1.)$ $a$ has infinite order (Exercise 14.2, Step 1), so $\langle a \rangle \cong \mathbb{Z}$ is free on $\{a\}$.

**Step 2.** $(n \geq 2.)$ Let $k = n - 1$. Lemma 11.18 gives an onto $\phi : F \to \mathbb{Z}/k$, $a \mapsto 1$, $b \mapsto 0$, so $H = \ker \phi$ has index $k$.

**Step 3.** By Theorem 14.9, $H$ is free of rank $k(2-1)+1 = n$. ■

**Exercise 14.4 (subgroups of index 2)**

Let $F$ be free of rank $n \geq 1$. Show that $F$ has exactly $2^n - 1$ subgroups of index 2, and that each is free of rank $2n - 1$.

**Solution [Supplement].**

**Step 1.** Index 2 $\Rightarrow$ normal ($gH = Hg = F - H$ for $g \notin H$), so $H = \ker(F \to F/H \cong \mathbb{Z}/2)$; conversely kernels of surjections $F \to \mathbb{Z}/2$ have index 2.

**Step 2.** A surjection $\phi : F \to \mathbb{Z}/2$ is determined by its kernel ($\phi = 0$ on $H$, 1 off $H$), so index-2 subgroups $\leftrightarrow$ surjections $F \to \mathbb{Z}/2$.

**Step 3.** By Lemma 11.18, homomorphisms $F \to \mathbb{Z}/2 \leftrightarrow$ maps $\{x_1, \ldots, x_n\} \to \mathbb{Z}/2$: $2^n$ of them, all onto except 0. So there are $2^n - 1$.

**Step 4.** By Theorem 14.9, each has rank $2(n-1)+1 = 2n-1$. ■